4 Number Sense

Big idea. "Number sense" is the ability to *understand* numbers, not just to *compute* with them: knowing which number is larger, whether a result is reasonable, how to round for a quick estimate, and seeing the hidden structure inside a number (its digits, its divisors, its multiples). In this topic you will review and connect:

- **decimals and negative numbers**: comparing and ordering them on the number line;
- the **order of operations** and **estimating** results;
- **fractions, ratios, percentages**: "a fraction of a number", equivalent fractions, discounts;
- **speed** – distance – time and unit conversion (km/h, m/s);
- **divisibility, common divisors, last digits, place value** (solving digit puzzles);
- the **mean, the median** and **number patterns** (Fibonacci, arithmetic sequences).

The general secret: *estimate before you calculate; check after you calculate.* A student with good number sense can often eliminate several unreasonable options in a multiple-choice question without any exact calculation.

① Decimals: reading, comparing, ordering, adding and subtracting

1. Place value. In a decimal, each digit sits in a *place*. After the decimal point come the tenths, the hundredths, the thousandths, ...

Tens	Ones	.	Tenths	Hundredths	Thousandths
	2	.	2	3	
	2	.	3	0	

Example: $2.23 = 2 + \frac{2}{10} + \frac{3}{100}$, while $2.3 = 2 + \frac{3}{10} = 2.30$.

2. Adding or removing zeros at the end of the decimal part does not change the value: $2.3 = 2.30 = 2.300$. Reason: $\frac{3}{10} = \frac{30}{100} = \frac{300}{1000}$ (multiply numerator and denominator by 10).

3. Rule for comparing two positive decimals.

- Compare the *whole-number parts* first: the one with the larger whole part is larger.
- If the whole parts are equal, *add zeros* so that both decimal parts have the same number of digits, then compare place by place from left to right: tenths, then hundredths, ...

Example: compare 0.304 and 0.34. Write $0.34 = 0.340$. Tenths: $3 = 3$; hundredths: $0 < 4$. So $0.304 < 0.340 = 0.34$.

4. Adding and subtracting decimals: write the numbers in a column *with the decimal points lined up* (places lined up), then add/subtract as with whole numbers.

5. Rounding decimals. To round to the nearest hundredth (two digits after the point), look at the digit just to the right of that place (the thousandths digit): if it is less than 5, keep the

hundredths digit; if it is 5 or more, add 1 to the hundredths digit; then drop all digits after it. Example: $4.6875 \approx 4.69$ (the thousandths digit is $7 \geq 5$), while $3.1416 \approx 3.14$ (because $1 < 5$). The sign "$\approx$" reads "is approximately".

Why is it true? Why can't we compare "by length" (the number with more digits is larger)? Write both in thousandths: $0.304 = \frac{304}{1000}$ and $0.34 = \frac{340}{1000}$. The fractions have the same denominator and $304 < 340$, so $0.304 < 0.34$, even though 0.304 "looks longer". Adding zeros is exactly bringing both to the same denominator. □

Common mistake. Many students think $2.23 > 2.3$ because "$23 > 3$". Wrong! Compare place by place: 2.23 has 2 tenths, while 2.3 has 3 tenths, so $2.23 < 2.3$.

ThinkAlgo Tip. To find the smallest (largest) number in a set of decimals, write all of them in *one column* with the decimal points lined up, adding zeros to make them equally long. Then you just compare them like whole numbers.

② Negative numbers and the number line

1. The number line. The integers are placed on a straight line: 0 in the middle, positive numbers to the right, negative numbers to the left.



2. Comparison rule. On a (horizontal) number line, *a number to the left is smaller than a number to the right.* Therefore:

- every negative number is less than 0, and 0 is less than every positive number;
- every negative number is less than every positive number;
- of two negative numbers, the one *farther from* 0 is *smaller*: $-8 < -2$ because $8 > 2$.

3. Everyday pictures. A temperature of $-8°C$ is colder than $-2°C$; owing 8 thousand dong leaves you "poorer" than owing 2 thousand dong.

4. The coordinate plane. A point is given by a pair of numbers $(x; y)$: x is the horizontal coordinate (moving right/left), y is the vertical coordinate (moving up/down). A horizontal move changes only x; a vertical move changes only y.

Why is it true? Why is $-8 < -2$? We have $-2 = -8 + 6$: to go from -8 to -2 on the number line we must move 6 units *to the right*. Moving right means increasing, so -2 is larger than -8. □

Common mistake. Do not compare negative numbers "by the feel of positive numbers". -8 is *less* than -2, even though 8 is greater than 2. When in doubt, sketch a quick number line.

ThinkAlgo Tip. **Parity (even − odd)** is a powerful "number sense" tool: even + even = even, odd + odd = even, even + odd = odd; a product of odd numbers is always odd. For instance, the

sum of 5 odd numbers is odd, and the sum of 4 odd numbers is even. When a point moves on a coordinate grid, looking at the parity of its coordinates often tells you at once whether a position can be reached.

③ Order of operations and estimation

1. Order of operations.

1. *Brackets* first: ( ), then [ ], then { }.
2. Next, *powers*.
3. Then *multiplication and division*, from left to right.
4. Finally *addition and subtraction*, from left to right.

Example: $6\times2-3=12-3=9$ (multiply first, then subtract), whereas $6\times(2-3)=6\times(-1)=-6$.
2. Rounding to estimate. To know quickly "roughly how much" a result is, round the factors to the nearest ten or hundred and compute mentally. Example: $99\times9\approx100\times9=900$.
3. A quick-calculation trick: $99\times9=(100-1)\times9=900-9=891$ (multiplication distributes over subtraction).

Why is it true? Why is $(100-1)\times9=100\times9-1\times9$? Imagine 9 rows of 100 sweets each, but 1 sweet is taken from every row. The sweets left are 9 rows × 99 sweets, which also equals (9 × 100 sweets) minus (the 9 sweets taken). This is the *distributive property*: $a\times(b-c)=a\times b-a\times c$.□

Common mistake. Working "from left to right regardless of the operation" is wrong: $2+3\times4\neq5\times4$. The correct value is $2+3\times4=2+12=14$.

ThinkAlgo Tip. In a multiple-choice question asking "which number is closest to…", the options often differ by factors of 10 (10; 100; 1000; …). Then a rough estimate is enough to choose correctly.

④ Fractions: a fraction of a number, equivalent fractions

1. A fraction is a part of a whole. $\frac{a}{b}$ means: split the whole into b equal parts and take a of them.
2. Finding a fraction of a number: to find $\frac{m}{n}$ of a, compute $a\times\frac{m}{n}$ (split a into n parts and take m of them). Example: $\frac{1}{3}$ of 396 is $396\div3=132$.
3. The part that is left. If $\frac{m}{n}$ of a job is done, the part left is $1-\frac{m}{n}=\frac{n-m}{n}$.
4. Equivalent fractions (the basic property). Multiplying (or dividing) the numerator and the denominator by the same non-zero number gives an equal fraction:

$$\frac{3}{7}=\frac{3\times9}{7\times9}=\frac{27}{63},\qquad\frac{12}{36}=\frac{12\div12}{36\div12}=\frac{1}{3}.$$

5. Simplifying a fraction: divide the numerator and the denominator by their greatest common

divisor to get a fraction in lowest terms.

6. "Sum and ratio" problems. If two numbers are in the ratio $\frac{5}{8}$, we can think of the first number as 5 equal parts and the second as 8 such parts. Knowing the sum (or the difference) gives the value of one part.

7. A fraction of a fraction. "Half of a quarter" is $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$: multiply numerator by numerator, denominator by denominator. This happens often when a shape is halved several times in a row: each halving multiplies the area fraction by $\frac{1}{2}$ again.

8. Comparing fractions. Bring them to a common denominator and compare numerators, or convert them to decimals. Examples: $\frac{2}{3} = \frac{8}{12} > \frac{7}{12}$; $\frac{3}{5} = 0.6 < \frac{3}{4} = 0.75$.



Why is it true? Why does multiplying the numerator and denominator by the same number leave the fraction unchanged? $\frac{3}{7}$ is "cut the cake into 7 parts, take 3". If we cut *each* part into 9 smaller pieces, the cake has $7 \times 9 = 63$ pieces and our share has $3 \times 9 = 27$ pieces. The amount of cake has not changed, so $\frac{3}{7} = \frac{27}{63}$. □

Common mistake. Read the question carefully: does it ask for the part *done* or the part *left*? Many students correctly find $\frac{1}{3}$ of 396 to be 132 and circle it straight away, while the question asks for the pages *left*, $396 - 132 = 264$.

Note for parents. With fractions, a picture beats any explanation. Draw a strip (a chocolate bar) split into equal parts and shade the part used; your child will see the part left by themselves. This is the familiar "bar model" from primary school.

⑤ Percentages, unit price and discounts

1. Percent means "out of one hundred". $80\% = \frac{80}{100} = 0.8$.

2. Finding $p\%$ of a number a: compute $a \times \frac{p}{100}$. Example: 80% of 50 is $50 \times \frac{80}{100} = 40$.

3. Expressing a as a percentage of b: compute $\frac{a}{b} \times 100\%$. Example: 32 correct answers out of 40 questions is $\frac{32}{40} \times 100\% = 80\%$.

4. Unit price. The unit price is the price of *one* unit (one kilogram, one book, ...):

$$\text{Total cost} = \text{unit price} \times \text{quantity}, \qquad \text{unit price} = \text{total cost} \div \text{quantity}.$$

5. A discount of $p\%$. New price = old price $\times \left(1 - \frac{p}{100}\right)$. A 10% discount leaves 90%: multiply by 0.9.

6. Successive discounts. Each time the price is reduced by 10% *of the current price*, we multiply by 0.9 once more. Example: a price of 100 thousand dong after two reductions: $100 \to 90 \to 81$ (thousand dong), *not* 80.

Why is it true? Why are two 10% reductions not the same as one 20% reduction? The second reduction is 10% of 90 (that is 9), not 10% of 100 (that is 10). Since the price has already dropped, the next reduction is smaller too: in total the price drops by only $10 + 9 = 19$ thousand dong, i.e. 19%. □

ThinkAlgo Tip. Finding 10% very fast: just divide by 10 (move the decimal point one place to the left). From 10% you get 5% (halve it), 20% (double it), $80\% = 100\% - 20\%$, ...

⑥ Speed, distance, time and unit conversion

1. Three linked formulas. For motion at a constant speed:

$$s = v \times t, \qquad v = s \div t, \qquad t = s \div v,$$

where s is the distance, v the speed and t the time.

2. Who is faster? The faster person covers a *longer* distance in the *same amount of time*. The safest way: compute each person's speed in the same unit (for example m/s) and compare.

3. Running laps. Running n laps of a track L metres long means a distance of $n \times L$ metres.

4. Converting km/h to m/s. 1 km = 1000 m and 1 hour = 3600 seconds, so

$$1 \text{ km/h} = \frac{1000 \text{ m}}{3600 \text{ s}} = \frac{5}{18} \text{ m/s}, \qquad 18 \text{ km/h} = 18 \times \frac{5}{18} = 5 \text{ m/s}, \qquad 1 \text{ m/s} = 3.6 \text{ km/h}.$$

$s = v \times t$

Cover the quantity you want:
$v = s \div t$ and $t = s \div v$

Common mistake. The quantities must use *matching units*: if the distance is in metres and the time in seconds, the speed comes out in m/s; do not divide metres by hours. Before comparing two speeds, convert them to the same unit.

⑦ Divisibility, common divisors, last digits and place value

1. Multiples and divisors. a is a *multiple* of b (or b is a *divisor* of a) if $a = b \times k$ for some whole number k. Example: $45 = 9 \times 5$, so 45 is a multiple of 9.

2. Divisibility rules.

- by 2: the last digit is $0, 2, 4, 6, 8$; by 5: the last digit is 0 or 5; by 10: the last digit is 0;
- by 3: the digit sum is divisible by 3; by 9: *the digit sum is divisible by* 9.

3. Common divisors. A number that divides both a and b is a common divisor of a and b. Every

common divisor is a divisor of the *greatest common divisor* $\gcd(a, b)$. Example: $\gcd(40, 72) = 8$, so the common divisors of 40 and 72 are $1, 2, 4, 8$.

4. The last digit of a product depends only on the last digits of the factors: $\ldots 3 \times \ldots 7$ ends in the same digit as $3 \times 7 = 21$, namely 1.

5. Place value. A number written with digits can be expanded by place:

$$\overline{abc} = 100a + 10b + c, \qquad \overline{PQQ} = 100P + 10Q + Q = 100P + 11Q.$$

(The bar on top shows that this is the number with digits a, b, c, not the product $a \times b \times c$.)

Why is it true? Why does the divisibility rule for 9 work? Take the number $\overline{abc} = 100a+10b+c$ and split it:

$$100a + 10b + c = (99a + 9b) + (a + b + c) = 9 \times (11a + b) + (a + b + c).$$

The part $9 \times (11a + b)$ is always divisible by 9. So $\overline{abc}$ is divisible by 9 exactly when the remaining part $a + b + c$ (the digit sum) is divisible by 9. For numbers with more digits the argument is the same, because $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$, $\ldots$ □

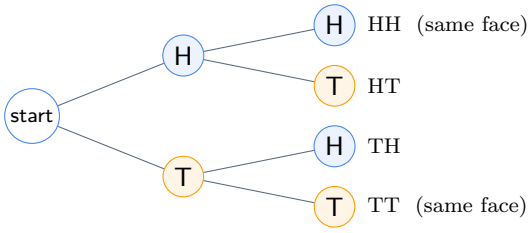
ThinkAlgo Tip. For a "digit puzzle" (each letter stands for a digit) such as $\overline{PQQ} + \overline{PPQ} + \overline{QQQ} = 876$, work *from the ones column to the left*, watching out for carries. The ones column usually has the fewest possibilities, so the first digit is found very quickly. You can also use place value to turn the puzzle into one equation. When you have to count or find all possibilities (sets of digits, outcomes of tossing coins, $\ldots$), *list them systematically* (in increasing order, or with a table/tree diagram) so that nothing is missed or repeated.

Systematic listing and chance.

- To count the three-digit numbers whose digits have a given *product*, find the sets of three digits (written in increasing order so as not to repeat), then count the arrangements of each set: three different digits can be arranged in $3 \times 2 \times 1 = 6$ ways; two equal digits and one other in 3 ways. Example: product 12 with the set $\{1; 3; 4\}$ gives 6 numbers $134, 143, 314, 341, 413, 431$; the set $\{2; 2; 3\}$ gives 3 numbers $223, 232, 322$.
- When all outcomes are *equally likely*, the probability of an event is

$$\frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Example: tossing 2 coins has 4 equally likely outcomes: HH, HT, TH, TT (H: heads, T: tails). The probability that both coins show the same face is $\frac{2}{4} = \frac{1}{2}$. Use a tree diagram to list them:



Common mistake. "The digit sum is divisible by 9" is the rule for 9; it does *not* work for 4, 6 or 8. And a number divisible by 3 is not necessarily divisible by 9: 12 is divisible by 3 but not by 9.

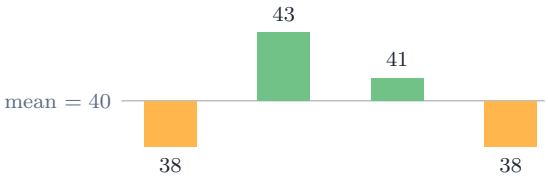
⑧ Number patterns, the mean and the median

1. The Fibonacci sequence. $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$ Each number (from the third on) is *the sum of the two numbers just before it*: $2 = 1 + 1$, $3 = 1 + 2$, $5 = 2 + 3$, ...
2. Arithmetic sequences (equally spaced). Each number (from the second on) equals the previous number plus a fixed number d (the common difference). Example: $1, 4, 7, 10$ (with $d = 3$). The n-th number is $a_1 + (n - 1) \times d$.
3. The mean (average). The mean of n numbers = (their sum) $\div n$. Hence

$$\text{Sum} = \text{mean} \times \text{how many numbers.}$$

4. The "deviation" method. The deviations from the mean always add up to 0: the "excess" of the numbers above the mean exactly equals the "shortfall" of the numbers below the mean.
5. The median. Arrange the numbers in increasing order; the number in the middle is the median (when there is an odd number of numbers). Example: the median of $42, 43, 46, 49, 50$ is 46. With an even number of numbers, the median is the mean of the two middle numbers: the median of $6, 7, 9, 10$ is $(7 + 9) \div 2 = 8$. Note: when a number is added or changed, the "middle" position can move, so re-order the list.
6. Consecutive integers. For an odd number of consecutive integers, the mean is the middle number: the mean of $19, 20, 21, 22, 23$ is 21.

Why is it true? Why do the deviations add up to 0? Suppose 4 numbers have mean 80, so their sum is $4 \times 80 = 320$. Write each number as 80+ (its deviation). Adding up: 320+ (sum of deviations) $= 320$, so the sum of the deviations must be 0. □



The four numbers $38, 43, 41, 38$ have mean 40. Deviations from 40: -2, $+3$, $+1$, -2; sum $= 0$.

Note for parents. You can illustrate the mean with stacks of coins: "level out" the tall stacks into the short ones until all stacks are the same height. That common height is the mean. The coins moved out of the tall stacks exactly equal the coins added to the short ones – this is the idea "the deviations add up to 0".

Worked examples

Example 1

Consider the set $\{0.5;\ 0.45;\ 0.405;\ 0.054;\ 0.54\}$.

a) Arrange the numbers in increasing order.

b) Find the sum of the largest and the smallest number.

Guided solution.

Step 1. Write the numbers in a column, adding zeros to get three digits after the point

Adding zeros at the end of the decimal part does not change the value, so:

$$0.5 = 0.500; \quad 0.45 = 0.450; \quad 0.405; \quad 0.054; \quad 0.54 = 0.540.$$

Step 2. Compare them like whole numbers

Dropping "0.", we compare 500; 450; 405; 54; 540:

$$54 < 405 < 450 < 500 < 540.$$

So the increasing order is: $0.054 < 0.405 < 0.45 < 0.5 < 0.54$.

Step 3. Add the largest and the smallest (decimal points lined up)

$$\begin{array}{r} 0.540 \\ +\ 0.054 \\ \hline 0.594 \end{array}$$

Check: $0.594 - 0.54 = 0.054$. Correct.

Answer: **a)** $0.054 < 0.405 < 0.45 < 0.5 < 0.54$; **b)** 0.594

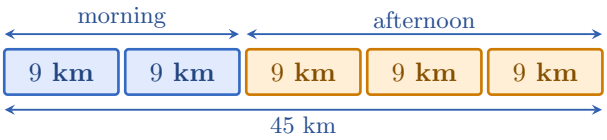
Example 2

Hoa cycles along a 45 km route. In the morning she covers $\frac{2}{5}$ of the route. How many kilometres does she still have to ride in the afternoon, and what fraction of the whole route is that?

Guided solution.

Step 1. Draw a bar model

Split the route into 5 equal parts; the morning covers 2 parts, the afternoon 3 parts.



Step 2. Find the value of one part

One part is $45 \div 5 = 9$ (km).

Step 3. Morning and afternoon distances

Morning: $2 \times 9 = 18$ (km). Afternoon: $45 - 18 = 27$ (km), which is also $3 \times 9 = 27$ (km).

Step 4. **The fraction left** $1 - \frac{2}{5} = \frac{5}{5} - \frac{2}{5} = \frac{3}{5}$. Check: $\frac{27}{45} = \frac{27 \div 9}{45 \div 9} = \frac{3}{5}$.

Answer: 27 **km, which is** $\frac{3}{5}$ **of the route**

Example 3

a) Evaluate $20 - 4 \times 3 + 8 \div 2$ and $(20 - 4) \times (3 + 8) \div 2$.
b) Without calculating exactly, decide which number 101×49 is closest to: 50; 500; 5000; 50 000? Then compute it exactly with a quick method.

Guided solution.

Step 1. **No brackets: multiply and divide first; add and subtract after**

$$20 - \underbrace{4 \times 3}_{12} + \underbrace{8 \div 2}_{4} = 20 - 12 + 4 = 8 + 4 = 12.$$

(Subtraction and addition are done from left to right: $20 - 12 = 8$, then $8 + 4 = 12$.)

Step 2. **With brackets: brackets first**

$$(20 - 4) \times (3 + 8) \div 2 = 16 \times 11 \div 2 = 176 \div 2 = 88.$$

Step 3. **Estimate**
$101 \approx 100$ and $49 \approx 50$, so $101 \times 49 \approx 100 \times 50 = 5000$. Choose 5000.

Step 4. **Exact value using the distributive property**
$101 \times 49 = (100 + 1) \times 49 = 4900 + 49 = 4949$, indeed very close to 5000.

Answer: **a)** 12 **and** 88; **b) closest to** 5000; $101 \times 49 = 4949$

Example 4

a) Rice costs 18 thousand dong per kilogram. Mum buys 15 kg and gets a 10% discount on the whole bill. How much does she pay?
b) A shirt costs 400 thousand dong. After every week the shop reduces the *current price* by 10%. After at least how many weeks is the price of the shirt below 300 thousand dong?

Guided solution.

Step 1. **Total before the discount**
Total cost = unit price × quantity = $18 \times 15 = 270$ (thousand dong).

Step 2. **Subtract** 10%
10% of 270 is $270 \div 10 = 27$ (thousand dong). Mum pays $270 - 27 = 243$ (thousand dong) = 243 000 dong. (Or: 90% remains, i.e. $270 \times 0.9 = 243$.)

Step 3. **Successive discounts: multiply by** 0.9 **every week**

Time	Start	After 1 week	After 2 weeks	After 3 weeks
Price (thousand dong)	400	360	324	291.6
Reduction	–	40	36	32.4

After 2 weeks the price is $324 > 300$; after 3 weeks it is $291.6 < 300$.

Common mistake. Do not think "the price drops by 40 thousand every week" (because 10% of 400 is 40). Each week the reduction is 10% of *the price at that time*, so the reductions get smaller and smaller: 40, then 36, then 32.4, ...

Answer: a) 243 000 dong; b) 3 weeks

Example 5

a) Bình cycles 3 laps around a stadium with a perimeter of 400 m in 4 minutes. An cycles at 15 km/h. Who is faster?
b) A car travels at 72 km/h. How many metres does it travel in 3 seconds?

Guided solution.

Step 1. Bình's speed (m/s)
Distance: $3 \times 400 = 1200$ (m). Time: 4 minutes $= 4 \times 60 = 240$ (seconds).

$$v_{\text{Bình}} = 1200 \div 240 = 5 \text{ (m/s)}.$$

Step 2. Convert An's speed to m/s

Recall. $1 \text{ km/h} = \dfrac{1000 \text{ m}}{3600 \text{ s}} = \dfrac{5}{18} \text{ m/s}.$

$v_{\text{An}} = 15 \times \dfrac{5}{18} = \dfrac{75}{18} \approx 4.17$ (m/s). Since $5 > 4.17$, **Bình is faster**. (Another way: 5 m/s $= 5 \times 3.6 = 18$ km/h > 15 km/h.)

Step 3. The car
72 km/h $= 72 \times \dfrac{5}{18} = 20$ m/s. In 3 seconds it travels $20 \times 3 = 60$ (m).

Answer: a) Bình is faster (5 m/s against about 4.17 m/s); b) 60 m

Example 6

a) Which of the numbers 72; 84; 117; 135 are divisible by 9?
b) Find the digit x such that the number $\overline{3x5}$ is divisible by 9.
c) How many whole numbers are divisors of both 24 and 36?

Guided solution.

Step 1. Use the rule for 9: compute the digit sum
72: $7 + 2 = 9$ (divisible by 9). 84: $8 + 4 = 12$ (not divisible by 9).
117: $1 + 1 + 7 = 9$ (divisible). 135: $1 + 3 + 5 = 9$ (divisible).
So 72, 117, 135 are divisible by 9; 84 is not ($84 = 9 \times 9 + 3$).

Step 2. Find the digit x
The digit sum of $\overline{3x5}$ is $3 + x + 5 = 8 + x$. Since x is a digit, $8 \le 8 + x \le 17$. In this range only 9 is divisible by 9, so $8 + x = 9$, i.e. $x = 1$. The number is $315 = 9 \times 35$.

Step 3. Common divisors of 24 and 36
Divisors of 24: $1, 2, 3, 4, 6, 8, 12, 24$. Divisors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$.
The numbers in both lists: $1, 2, 3, 4, 6, 12$ – these are exactly the divisors of $\gcd(24, 36) = 12$.

Answer: a) 72; 117; 135; b) $x = 1$; c) 6 numbers

Example 7

In the calculation $\overline{AB} + \overline{AB} + \overline{AB} = \overline{1AB}$, each letter stands for a digit. Find A and B.

Guided solution.

Method 1 – Place value. We have $\overline{1AB} = 100 + \overline{AB}$. So

$$3 \times \overline{AB} = 100 + \overline{AB} \Rightarrow 2 \times \overline{AB} = 100 \Rightarrow \overline{AB} = 50.$$

So $A = 5$, $B = 0$.

Method 2 – Column by column. Ones column: $B + B + B = 3B$ ends in the digit B, so $3B - B = 2B$ ends in 0, hence $B = 0$ or $B = 5$.

- If $B = 5$: $3 \times 5 = 15$, write 5 carry 1. Tens column: $3A + 1 = \overline{1A} = 10 + A$, so $2A = 9$ – no digit works. Rejected.
- If $B = 0$: no carry. Tens column: $3A = 10 + A$, so $2A = 10$, $A = 5$.

Check: $50 + 50 + 50 = 150 = \overline{1AB}$ with $A = 5$, $B = 0$. Correct.

Answer: $A = 5$, $B = 0$

Example 8

a) Nam's four test scores are 6; 9; 10 and x. The mean of the four scores is 8. Find x.
b) The mean of five consecutive whole numbers is 30. Find the smallest of them.

Guided solution.

Step 1. Using the sum

Recall. Sum = mean × how many numbers.

The four scores add up to $8 \times 4 = 32$. So $x = 32 - 6 - 9 - 10 = 7$.

Step 2. Using deviations (faster)

Compared with 8: the 6 is 2 short, the 9 is 1 over, the 10 is 2 over. Sum of deviations: $-2+1+2 = +1$. For all deviations to add up to 0, the fourth score must be 1 short: $x = 8-1 = 7$.

Step 3. Five consecutive numbers

For five consecutive numbers, the mean is the middle number. The middle number is 30, so the five numbers are $28, 29, 30, 31, 32$. The smallest is 28.

Check: $28 + 29 + 30 + 31 + 32 = 150 = 5 \times 30$. Correct.

Answer: a) $x = 7$; b) 28

Core practice

Problem 1

The smallest number in the set $\{3.2;\ 2.3;\ 3;\ 2.23;\ 3.22\}$ is

(A) 3.2 (B) 2.3 (C) 3 (D) 2.23 (E) 3.22

Detailed solution

Understanding the problem. There are 5 decimals. We need the *smallest* one.

> **Recall.** Comparing decimals: compare the whole parts first; if they are equal, compare the tenths, then the hundredths, ... You may add zeros at the end of the decimal part without changing the value.

Step 1. Compare the whole parts

Whole parts: $3.2 \to 3$; $2.3 \to 2$; $3 \to 3$; $2.23 \to 2$; $3.22 \to 3$.

The smallest number must have the smallest whole part, namely 2. Only two candidates remain: 2.3 and 2.23.

Step 2. Use the same number of decimal places and compare

$2.3 = 2.30$ and 2.23. Tenths: 3 against 2; since $2 < 3$, we get $2.23 < 2.30$.

Step 3. Conclusion

Ordering everything to be sure: $2.23 < 2.3 < 3 < 3.2 < 3.22$. The smallest number is 2.23.

> **Common mistake.** Do not choose 2.3 just because "3 is less than 23". The digit 3 in 2.3 means 3 tenths, while 2.23 has only 2 tenths.

Check: $2.23 = \frac{223}{100}$ and $2.3 = \frac{230}{100}$; $223 < 230$.

The number 2.23 is option **(D)**.

Answer: (D) 2.23

Problem 2

Minh goes on a 36 km kayak trip. Before lunch he paddles the first 12 km. After lunch, what fraction of the whole trip does Minh still have to paddle?

(A) $\frac{1}{2}$ (B) $\frac{5}{6}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{5}$

Detailed solution

Understanding the problem. The whole trip is 36 km and 12 km are done. We are asked for the part *left* (not yet paddled), as a fraction.

Step 1. Distance left

$36 - 12 = 24$ (km).

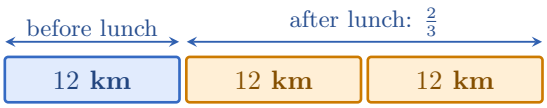
Step 2. Form the fraction "part left over the whole"

Recall. The fraction "what part of the whole" $= \dfrac{\text{the part}}{\text{the whole}}$. Simplify by dividing numerator and denominator by their greatest common divisor.

$$\frac{24}{36} = \frac{24 \div 12}{36 \div 12} = \frac{2}{3}.$$

Step 3. Picture it

Split the 36 km into 3 sections of 12 km: Minh has done 1 section, 2 sections are left.



Common mistake. The fraction $\dfrac{12}{36} = \dfrac{1}{3}$ is the part *done*. The question asks for the part *left*, so the answer is $1 - \dfrac{1}{3} = \dfrac{2}{3}$.

Check: $\dfrac{2}{3} \times 36 = 24$ km; $24 + 12 = 36$ km. Correct.

Note for parents. Ask your child two questions: "What part is done?" and "What part is left?". Getting into the habit of answering both helps avoid the trap of choosing $\dfrac{1}{3}$.

The fraction $\dfrac{2}{3}$ is option **(D)**.

Answer: (D) $\dfrac{2}{3}$

Problem 3

Which of the following statements is **true**?

(A) 0 is less than -5 (B) 7 is less than -1 (C) 10 is less than $\dfrac{1}{4}$

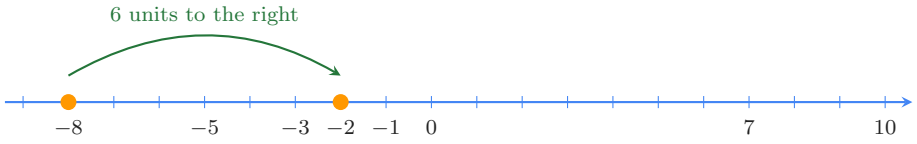
(D) -1 is less than -3 (E) -8 is less than -2

Detailed solution

Understanding the problem. Each option is a comparison. We check them one by one to find the true one.

Recall. On a horizontal number line, a number to the left is smaller than a number to the right. Every negative number is less than 0 and less than every positive number. Of two negative numbers, the one farther from 0 is smaller.

Step 1. Place the numbers on a number line



Step 2. Check each option

(A) 0 is less than -5? False: -5 is to the left of 0, so $-5 < 0$.
(B) 7 is less than -1? False: a positive number is always greater than a negative number.
(C) 10 is less than $\frac{1}{4}$? False: $\frac{1}{4} = 0.25 < 10$.
(D) -1 is less than -3? False: -3 is farther from 0, so $-3 < -1$.
(E) -8 is less than -2? **True**: -8 is to the left of -2 (farther from 0).

Common mistake. Many students choose (D), thinking "$1 < 3$ so $-1 < -3$". For negative numbers it is the other way round: $-1 > -3$ (a temperature of $-1°$C is warmer than $-3°$C).

Check: $-2 - (-8) = 6 > 0$, which means -2 is greater than -8.
The only true statement is **(E)**.

Answer: (E) -8 is less than -2

Problem 4

The value of $6 \times 2 - 3$ is

(A) 9 (B) -6 (C) 12 (D) 15 (E) 10

Detailed solution

Understanding the problem. The expression has two operations, multiplication and subtraction, and no brackets.

Recall. In an expression without brackets: multiply and divide first; add and subtract after.

Step 1. Do the multiplication first
$6 \times 2 = 12$.

Step 2. Then the subtraction
$12 - 3 = 9$.

Common mistake. Subtracting first would give $6 \times (2 - 3) = 6 \times (-1) = -6$ – option (B), a "trap" for anyone who forgets the order of operations. The expression has *no brackets*, so we must multiply first.

Check: Read the expression as a story: "6 boxes with 2 cakes each, 3 cakes eaten" leaves $12 - 3 = 9$ cakes.
The value 9 is option **(A)**.

Answer: (A) 9

Problem 5

A test is marked out of 50. Mai scores 80%. How many marks does Mai get?

(A) 40 (B) 62.5 (C) 10 (D) 45 (E) 35

Detailed solution

Understanding the problem. The maximum is 50 marks; Mai gets 80% of 50. We need Mai's marks.

Recall. To find p% of a number a, compute $a \times \frac{p}{100}$.

Step 1. Write the calculation
80% of 50 is $50 \times \frac{80}{100}$.

Step 2. Calculate
$50 \times \frac{80}{100} = \frac{50 \times 80}{100} = \frac{4000}{100} = 40$.

Step 3. A quick mental method
10% of 50 is 5; so $80\% = 8 \times 10\%$ is $8 \times 5 = 40$.

Common mistake. Option (B) 62.5 comes from dividing $50 \div 0.8$ – wrong, since 62.5 is even more than the maximum 50! Option (C) 10 is the number of marks Mai *lost* (20% of 50).

Check: $\frac{40}{50} = \frac{80}{100} = 80\%$. Correct.
The value 40 is option **(A)**.

Answer: (A) 40

Problem 6

Minced beef sells for 120 000 dong per kilogram. How much do 12 kg of minced beef cost?

(A) 120 000 dong (B) 12 000 dong (C) 1 440 000 dong
(D) 132 000 dong (E) 10 000 dong

Detailed solution

Understanding the problem. The unit price (price of 1 kg) is 120 000 dong; the quantity is 12 kg. We need the total cost.

> **Recall.** Total cost = unit price × quantity.

Step 1. Write the calculation
Total cost = $120\,000 \times 12$ (dong).

Step 2. Calculate (split $12 = 10 + 2$)
$120\,000 \times 10 = 1\,200\,000$; $120\,000 \times 2 = 240\,000$.
$120\,000 \times 12 = 1\,200\,000 + 240\,000 = 1\,440\,000$ (dong).

Step 3. Estimate to check
12 kg at more than 100 thousand dong per kg must cost more than 1 200 000 dong. Only option (C) fits.

> **Common mistake.** The wrong options come from using the wrong operation: (A) is the price of 1 kg; (D) is $120\,000 + 12\,000$ (adding instead of multiplying); (E) is $120\,000 \div 12$ (dividing instead of multiplying).

Check: $1\,440\,000 \div 12 = 120\,000$ dong per kg. Correct.
The amount 1 440 000 dong is option **(C)**.

Answer: (C) 1 440 000 dong

Problem 7

The value of $(8 \times 4) + 3$ is

(A) 96 (B) 15 (C) 56 (D) 35 (E) 28

Detailed solution

Understanding the problem. The brackets contain a multiplication; then 3 is added.

> **Recall.** Expressions with brackets: work out the brackets first.

Step 1. Work out the brackets
$8 \times 4 = 32$.

Step 2. Add 3
$32 + 3 = 35$.

> **Common mistake.** Misreading it as $8 \times (4 + 3) = 8 \times 7 = 56$ leads to option (C). The brackets surround 8×4, not $4 + 3$. Option (A) $96 = 8 \times 4 \times 3$ comes from mistaking the plus sign for a times sign.

Check: Without brackets the result is still $8 \times 4 + 3 = 35$, because multiplication always comes before addition; here the brackets only add emphasis.
The value 35 is option **(D)**.

Answer: (D) 35

Problem 8

Which of the following integers is closest to 99×9?

(A) 10 000 (B) 100 (C) 100 000 (D) 1000 (E) 10

Detailed solution

Understanding the problem. The options are 10; 100; 1000; 10 000; 100 000 – each 10 times the previous. An estimate of 99×9 is enough, but we also compute it exactly to be sure.

Step 1. Estimate

$99 \approx 100$, so $99 \times 9 \approx 100 \times 9 = 900$.

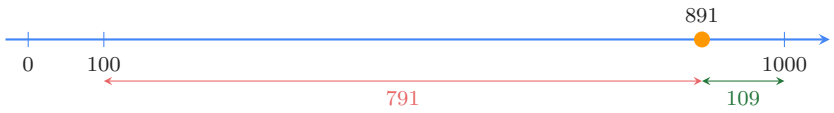
Step 2. Exact value using the distributive property

Recall. Distributive property: $(a - b) \times c = a \times c - b \times c$.

$99 \times 9 = (100 - 1) \times 9 = 900 - 9 = 891$.

Step 3. Compare the distances to the nearest options

The distance from 891 to 1000 is $1000 - 891 = 109$.
The distance from 891 to 100 is $891 - 100 = 791$.
The other options (10; 10 000; 100 000) are much farther away. So 1000 is the closest.



Check: $891 \div 9 = 99$. Correct.
The number 1000 is option **(D)**.

Answer: (D) 1000

Problem 9

Huy is reading a 396-page book. He has read only the first $\frac{1}{3}$ of the book. How many pages does he still have to read to finish the book?

(A) 264 (B) 124 (C) 250 (D) 199 (E) 244

Detailed solution

Understanding the problem. The whole book: 396 pages. Read so far: $\frac{1}{3}$ of the book. Asked: the number of pages *left*.

Recall. To find $\frac{m}{n}$ of a number, divide it by n and multiply by m.

Step 1. Pages read
$\frac{1}{3}$ of 396 is $396 \div 3 = 132$ (pages).

Step 2. Pages left
$396 - 132 = 264$ (pages).

Step 3. Another way: compute the part left directly
The part left is $1 - \frac{1}{3} = \frac{2}{3}$ of the book, i.e. $396 \div 3 \times 2 = 132 \times 2 = 264$ (pages).



Common mistake. Stopping at 132 answers the wrong question (that is the number of pages *read*; and 132 is not even among the options). Always re-read the question before choosing an answer.

Check: $264 + 132 = 396$, and $132 \times 3 = 396$. Correct.
The number 264 is option **(A)**.

Answer: (A) 264

Problem 10

Which of the following numbers is a multiple of 9?

(A) 50 (B) 40 (C) 35 (D) 45 (E) 55

Detailed solution

Understanding the problem. We need the number divisible by 9 (a multiple of 9) among the five given numbers.

Recall. A number is divisible by 9 exactly when its digit sum is divisible by 9.

Step 1. Digit sum of each number

Number	50	40	35	45	55
Digit sum	5	4	8	9	10
Divisible by 9?	no	no	no	**yes**	no

Step 2. Check with the 9 times table
Multiples of 9 nearby: 36, 45, 54. Only $45 = 9 \times 5$ appears among the options.

Why is it true? Why does the rule work for a two-digit number $\overline{ab}$? We have $\overline{ab} = 10a + b = 9a + (a + b)$. The part $9a$ is divisible by 9, so $\overline{ab}$ is divisible by 9 exactly when $a + b$ is. For 45: $45 = 9 \times 4 + 9$. □

Check: $45 \div 9 = 5$, remainder 0.
The number 45 is option **(D)**.

Answer: (D) 45

Problem 11

Consider the set $\{0.34;\ 0.304;\ 0.034;\ 0.43\}$. The sum of the smallest and the largest numbers in the set is

(A) 0.77 (B) 0.734 (C) 0.077 (D) 0.464 (E) 0.338

Detailed solution

Understanding the problem. Part 1: find the smallest and the largest number. Part 2: add them.

Recall. To compare decimals, add zeros so the decimal parts have equal length, then compare place by place from left to right.

Step 1. Write the numbers with three decimal places

Original	Rewritten	Tenths	Hundredths
0.34	0.340	3	4
0.304	0.304	3	0
0.034	0.034	0	3
0.43	0.430	4	3

Step 2. Find the smallest and the largest
Comparing like whole numbers: $34 < 304 < 340 < 430$, so $0.034 < 0.304 < 0.34 < 0.43$.
Smallest: 0.034 (its tenths digit is 0). Largest: 0.43 (its tenths digit is 4).

Step 3. Add, with the decimal points lined up

$$\begin{array}{r} 0.430 \\ +\ 0.034 \\ \hline 0.464 \end{array}$$

Common mistake. Misaligning the columns (adding 43 and 34 "by the last digits") gives 0.77 or 0.077 – options (A) and (C). Keep the decimal points in line: 4 tenths plus 0 tenths, 3 hundredths plus 3 hundredths, ...

Check: $0.464 - 0.43 = 0.034$. Correct.
The sum 0.464 is option **(D)**.

Answer: (D) 0.464

Problem 12

Lan and Hà run on a circular track 150 m long. Lan runs 3 laps in 120 seconds, and Hà runs 5 laps in 160 seconds. Who is the faster runner, and approximately what is her speed?

(A) Lan, 3.75 m/s (B) Lan, 2.4 m/s (C) Hà, 3.3 m/s

(D) Hà, 4.69 m/s (E) Hà, 3.75 m/s

Detailed solution

Understanding the problem. One lap is 150 m. Lan: 3 laps in 120 s. Hà: 5 laps in 160 s. We compute each girl's speed (m/s), compare, and choose the option.

Recall. Speed = distance ÷ time. The distance of n laps = $n\times$ the length of one lap.

Step 1. Distance run by each girl
Lan: $3 \times 150 = 450$ (m). Hà: $5 \times 150 = 750$ (m).

Step 2. Lan's speed
$v_{\text{Lan}} = 450 \div 120 = 3.75$ (m/s).

Step 3. Hà's speed
$v_{\text{Hà}} = 750 \div 160 = 4.6875 \approx 4.69$ (m/s).

Step 4. Compare
$4.69 > 3.75$, so **Hà is faster**, running at about 4.69 m/s.

	Distance	Time	Speed
Lan	450 m	120 s	3.75 m/s
Hà	750 m	160 s	$\approx$ 4.69 m/s

Common mistake. Option (E) "Hà, 3.75 m/s" pairs the right name with the slower girl's speed. Option (B) 2.4 m/s comes from the reversed calculation $120 \div 50$ (time divided by distance). Always write the units: m ÷ s = m/s.

Check: $4.6875 \times 160 = 750$ m = 5 laps; $3.75 \times 120 = 450$ m = 3 laps. Correct.
The conclusion "Hà, 4.69 m/s" is option **(D)**.

Answer: (D) Hà, 4.69 m/s

Problem 13

In the addition shown, P and Q each represent a digit. The value of $P + Q$ is

$$\begin{array}{r} P\,Q\,Q \\ P\,P\,Q \\ +\,Q\,Q\,Q \\ \hline 8\;7\;6 \end{array}$$

(A) 3 (B) 5 (C) 7 (D) 6 (E) 4

Detailed solution

Understanding the problem. The three numbers $\overline{PQQ}$, $\overline{PPQ}$, $\overline{QQQ}$ add up to 876. We need the digits P, Q, then $P + Q$. We go column by column, starting with the ones.

Step 1. Ones column: $Q + Q + Q = 3Q$ ends in 6
Since Q is a digit, $3Q$ is one of $0, 3, 6, \ldots, 27$. Of these, only 6 ends in 6 (16 and 26 are not multiples of 3). So $3Q = 6$, $Q = 2$, and there is *no carry* into the tens.

Step 2. Tens column: $Q + P + Q = P + 4$ ends in 7
$P + 4$ lies between 4 and 13; ending in 7 means $P + 4 = 7$, so $P = 3$. No carry into the hundreds.

Step 3. Hundreds column: check $P + P + Q = 8$
$3 + 3 + 2 = 8$. This matches the hundreds digit of 876.

> **Recall.** Place value: $\overline{PQQ} = 100P + 11Q$, $\overline{PPQ} = 110P + Q$, $\overline{QQQ} = 111Q$.

Step 4. Another way using place value
Adding: $(100P + 11Q) + (110P + Q) + 111Q = 210P + 123Q = 876$. Try $Q = 2$: $210P = 876 - 246 = 630$, so $P = 3$. (Other values of Q do not give a digit for P.)

Check: $322 + 332 + 222 = 876$. Correct.

> **Note for parents.** A digit puzzle is like a detective game: each column is a "clue". Let your child write each column as a sentence ("three times Q ends in 6") and try the digits 0 to 9. Do not forget to check the carries.

So $P + Q = 3 + 2 = 5$, which is option **(B)**.

Answer: (B) 5

Problem 14

The mean (average) of the four integers 78; 83; 82 and x is 80. Which of the following statements is true?

(A) x is 2 greater than the mean

(B) x is 1 less than the mean

(C) x is 2 less than the mean

(D) x is 3 less than the mean

(E) x is equal to the mean

Detailed solution

Understanding the problem. We know the mean of four numbers is 80 and we know three of them. Find x and compare it with 80.

Recall. Sum of the numbers = mean × how many numbers.

Step 1. Sum of the four numbers
$80 \times 4 = 320$.

Step 2. Find x
$78 + 83 + 82 = 243$, so $x = 320 - 243 = 77$.

Step 3. Compare with the mean
$80 - 77 = 3$: x is 3 less than the mean.

Step 4. A faster way – the deviation method
Compared with 80: 78 is 2 short; 83 is 3 over; 82 is 2 over. Sum of these three deviations: $-2 + 3 + 2 = +3$. Since all the deviations must add up to 0, x must be 3 "short": $x = 80 - 3 = 77$.

Why is it true? Each number can be written as 80+(its deviation). The sum of the four numbers is 4×80+(sum of deviations). But the sum of the four numbers is exactly 4×80, so the sum of the deviations is 0. □

Check: $(78 + 83 + 82 + 77) \div 4 = 320 \div 4 = 80$. Correct.

Note for parents. The "deviation" method is great for working out average marks mentally in everyday life. You can ask your child: "Three tests scored 7, 9, 8; to average 8, what does the fourth test need?" (Answer: 8, because $-1 + 1 + 0 = 0$.)

The statement "x is 3 less than the mean" is option **(D)**.

Answer: (D)

Problem 15

In the Fibonacci sequence $1, 1, 2, 3, 5, \ldots$, each number from the 2 onwards is the sum of the two numbers just before it. For example, the next number in the sequence is $3 + 5 = 8$. Which of the following numbers is in the sequence?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Detailed solution

Understanding the problem. Rule: next number = sum of the two before it. We continue the sequence until it passes 24 (the largest option), then see which option appears.

Step 1. Continue the sequence

Position	1	2	3	4	5	6	7	8	9
Number	1	1	2	3	5	8	13	**21**	34

In detail: $5 + 8 = 13$; $8 + 13 = 21$; $13 + 21 = 34$.

Step 2. Compare with the options

34 is already greater than 24, so all later numbers are greater than 24. Between 13 and 34 there is exactly one number of the sequence, 21. So 20, 22, 23, 24 are not in the sequence; only 21 is.

ThinkAlgo Tip. The Fibonacci sequence keeps increasing (from the second number on) because we keep adding positive numbers. So once the sequence has passed a number, that number can never appear later.

Check: $8 + 13 = 21$. Correct.
The number 21 is option **(B)**.

Answer: (B) 21

Problem 16

The mean (average) of five consecutive integers is 21. The smallest of the five integers is

(A) 17 (B) 21 (C) 1 (D) 18 (E) 19

Detailed solution

Understanding the problem. We have *five consecutive integers* (each one is exactly 1 more than the one before). Their mean is 21. We must find the *smallest* one.

Recall. Mean of a group of numbers = (sum of the numbers) ÷ (how many numbers there are). So: sum = mean × how many numbers.

Step 1. Find the sum of the five numbers

The sum of the five numbers is $21 \times 5 = 105$.

Step 2. Name the five numbers using the middle one

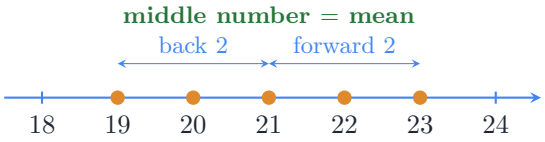
Call the middle (third) number n. Then the five consecutive numbers are

$$n - 2, \quad n - 1, \quad n, \quad n + 1, \quad n + 2.$$

Adding: $(n - 2) + (n - 1) + n + (n + 1) + (n + 2) = 5n$ (the -2 and $+2$, and the -1 and $+1$, cancel each other).

Step 3. Find the middle number, then the smallest

We have $5n = 105$, so $n = 105 \div 5 = 21$. The five numbers are 19, 20, 21, 22, 23. The smallest is $n - 2 = 19$.



Why is it true? With an *odd* number of consecutive integers, the numbers placed symmetrically around the middle one "balance out" (one is 1 less, one is 1 more; one is 2 less, one is 2 more...). So the sum equals (middle number) × (how many numbers), which means the mean is exactly the middle number. □

Common mistake. Do not choose 21 straight away: that is the *middle* number, not the smallest. And do not take $21 - 4 = 17$ either: from the middle number to the smallest is only 2 steps back, not 4.

Check: $19 + 20 + 21 + 22 + 23 = 105$ and $105 \div 5 = 21$. Correct.
The smallest number is 19, which is option **(E)**.

Answer: (E) 19

Problem 17

How many positive whole numbers, including 1, divide exactly into both 40 and 72?

(A) 9 (B) 12 (C) 4 (D) 2 (E) 5

Detailed solution

Understanding the problem. A number "divides exactly into both 40 and 72" when 40 is divisible by it *and* 72 is divisible by it. These are the *common divisors* (common factors) of 40 and 72. We must count them.

Recall. d is a divisor of a if a is divisible by d. A common divisor of two numbers is a divisor of both. The greatest common divisor (GCD) is the largest of the common divisors.

Step 1. List the divisors of 40
Find them in pairs whose product is 40: 1×40, 2×20, 4×10, 5×8. So the divisors of 40 are

$$\{1,\ 2,\ 4,\ 5,\ 8,\ 10,\ 20,\ 40\}.$$

Step 2. List the divisors of 72
Pairs with product 72: 1×72, 2×36, 3×24, 4×18, 6×12, 8×9. So the divisors of 72 are

$$\{1,\ 2,\ 3,\ 4,\ 6,\ 8,\ 9,\ 12,\ 18,\ 24,\ 36,\ 72\}.$$

Step 3. Pick the numbers that appear in both lists
Comparing the lists, the common numbers are 1, 2, 4, 8. So there are 4 common divisors.

ThinkAlgo Tip. A faster way: the common divisors of two numbers are exactly the divisors of their GCD. We have $40 = 2^3 \cdot 5$ and $72 = 2^3 \cdot 3^2$, so $\text{GCD}(40, 72) = 2^3 = 8$. The divisors of 8 are $1, 2, 4, 8$: exactly 4 numbers.

Why is it true? If d divides both 40 and 72, then d also divides the difference $72 - 40 = 32$, and then $40 - 32 = 8$. Conversely, every divisor of 8 divides $40 = 8 \cdot 5$ and $72 = 8 \cdot 9$. So "common divisors of 40 and 72" and "divisors of 8" are the same set. □

Common mistake. Option 12 is the number of divisors of 72 alone; options 9 or 5 usually come from missing or miscounting some. The question asks for *common* divisors.

Check: $40 \div 1$, $40 \div 2 = 20$, $40 \div 4 = 10$, $40 \div 8 = 5$; $72 \div 1$, $72 \div 2 = 36$, $72 \div 4 = 18$, $72 \div 8 = 9$ – all exact. But $3, 6, 9$ do not divide 40, and $5, 10$ do not divide 72.
There are 4 such numbers, which is option **(C)**.

Answer: (C) 4

Problem 18

The number that goes into the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Detailed solution

Understanding the problem. The two fractions are equal; the denominator has changed from 7 to 63. We need the new numerator.

Recall. Basic property of fractions: multiplying both the numerator and the denominator by the same non-zero number gives an equal fraction: $\frac{a}{b} = \frac{a \cdot m}{b \cdot m}$ $(m \neq 0)$.

Step 1. What was the denominator multiplied by?
$63 \div 7 = 9$, so the denominator 7 was multiplied by 9 to become 63.

Step 2. Multiply the numerator by the same number
To keep the fraction the same, the numerator must also be multiplied by 9: $3 \times 9 = 27$. So

$$\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}.$$

Common mistake. Option 9 is the *multiplier*, not the new numerator. Option 59 comes from the "adding" mistake: $63 - 7 = 56$ and then $3 + 56 = 59$ – adding the same number to the numerator and denominator does *not* keep the fraction the same.

Check: Cross-multiply: $3 \times 63 = 189$ and $7 \times 27 = 189$. The products are equal, so the fractions are equal.

The missing number is 27, which is option **(A)**.

Answer: (A) 27

Problem 19

A fraction is equal to $\frac{5}{8}$. Its numerator and denominator add up to 91. What is the difference between the denominator and the numerator of this fraction?

(A) 21 (B) 3 (C) 33 (D) 13 (E) 19

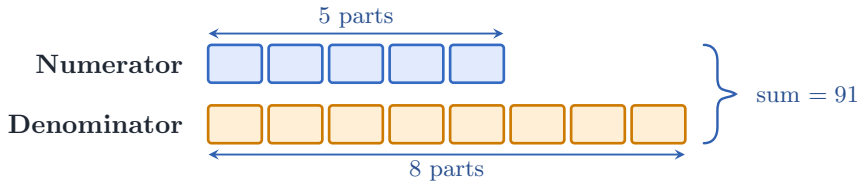
Detailed solution

Understanding the problem. The fraction equals $\frac{5}{8}$, so its numerator and denominator "share" in the ratio $5 : 8$. We know numerator + denominator $= 91$. We want denominator − numerator.

> **Recall.** Every fraction equal to $\frac{5}{8}$ has the form $\frac{5k}{8k}$ for some positive integer k (multiply the numerator and denominator by k).

Step 1. Draw a bar model

Think of the numerator as 5 equal parts and the denominator as 8 such parts.



Step 2. Find the value of one part

There are $5 + 8 = 13$ parts in total, worth 91. One part is $91 \div 13 = 7$.

Step 3. Find the numerator, the denominator and the difference

Numerator $= 5 \times 7 = 35$; denominator $= 8 \times 7 = 56$. The fraction is $\frac{35}{56}$.

Denominator − numerator $= 56 - 35 = 21$ (which is also 3 parts: $3 \times 7 = 21$).

> **Common mistake.** Option 3 is the difference $8 - 5$ of the original fraction $\frac{5}{8}$, not yet multiplied by the value of one part. Option 13 is the number of parts, not the difference.

Check: $35 + 56 = 91$; simplifying, $\frac{35}{56} = \frac{35 \div 7}{56 \div 7} = \frac{5}{8}$. Correct.

The difference is 21, which is option **(A)**.

Answer: (A) 21

Problem 20

Lan is driving at 80 km/h along a road with equally spaced power poles. Her car passes a pole every 4 seconds. Which of the following is closest to the distance between two neighbouring poles?

(A) 50 m (B) 60 m (C) 70 m (D) 80 m (E) 90 m

Detailed solution

Understanding the problem. In 4 seconds the car travels exactly the distance between two neighbouring poles. So we need the distance travelled in 4 seconds at 80 km/h. Careful: the speed is in *km and hours*, the time is in *seconds*, and the answers are in *metres* – we must convert units.

Recall. Distance = speed × time. 1 km = 1000 m; 1 hour = 60 minutes = 3600 seconds.

Step 1. Convert the speed to metres per second
80 km/h means 80 000 m in 3600 seconds. In 1 second the car travels

$$80\,000 \div 3600 = \frac{80\,000}{3600} = \frac{200}{9} \approx 22.2 \text{ m}.$$

Step 2. Distance travelled in 4 seconds

$$\frac{200}{9} \times 4 = \frac{800}{9} \approx 88.9 \text{ m}.$$

Step 3. Compare with the options
88.9 m is about 1.1 m from 90 m and about 8.9 m from 80 m. So the closest is 90 m.

ThinkAlgo Tip. Another way to think: in 1 hour = 3600 seconds the car passes $3600 \div 4 = 900$ gaps between poles. These 900 gaps make up 80 000 m altogether, so each gap is $80\,000 \div 900 \approx 88.9$ m.

Common mistake. If you forget to convert and compute $80 \times 4 = 320$, no option matches. Converting 1 hour as 60 seconds (wrong – it is 3600 seconds) also gives an absurd result.

Check: 90 m every 4 seconds is 22.5 m/s, i.e. $22.5 \times 3600 = 81\,000$ m per hour = 81 km/h, very close to 80 km/h. Reasonable.
Choose option **(E)**.

Answer: (E) 90 m

Problem 21

Each of the integers 334 and 419 has digits whose product is 36 (since $3 \cdot 3 \cdot 4 = 36$ and $4 \cdot 1 \cdot 9 = 36$). How many 3-digit positive integers have digits whose product is 36?

(A) 21 (B) 15 (C) 18 (D) 24 (E) 12

Detailed solution

Understanding the problem. A 3-digit number $\overline{abc}$ has digit product $a \cdot b \cdot c = 36$. No digit can be 0 (a 0 would make the product 0). A tidy method: *first* find the sets of three digits (ignoring order) with product 36, *then* count how many numbers each set can make.

Step 1. Find every set of three digits (from 1 to 9) with product 36
List each set in order $a \le b \le c$ and look at the smallest digit a.

- $a = 1$: we need $b \cdot c = 36$ with $1 \le b \le c \le 9$: $36 = 4 \cdot 9 = 6 \cdot 6$ (while $1 \cdot 36$, $2 \cdot 18$, $3 \cdot 12$ use a number bigger than 9). We get $\{1, 4, 9\}$ and $\{1, 6, 6\}$.
- $a = 2$: we need $b \cdot c = 18$ with $2 \le b \le c \le 9$: $18 = 2 \cdot 9 = 3 \cdot 6$. We get $\{2, 2, 9\}$ and $\{2, 3, 6\}$.
- $a = 3$: we need $b \cdot c = 12$ with $3 \le b \le c$: $12 = 3 \cdot 4$. We get $\{3, 3, 4\}$.
- $a \ge 4$: then $a \cdot b \cdot c \ge 4 \cdot 4 \cdot 4 = 64 > 36$, no sets.

So there are exactly 5 sets: $\{1, 4, 9\}$, $\{1, 6, 6\}$, $\{2, 2, 9\}$, $\{2, 3, 6\}$, $\{3, 3, 4\}$.

Recall. Counting arrangements: three *different* digits can be arranged into $3 \cdot 2 \cdot 1 = 6$ numbers (hundreds digit: 3 choices, tens: 2 choices, units: 1 choice). Three digits with *two equal* can be arranged into only 3 numbers (we just choose the position of the odd one out).

Step 2. Count the numbers made from each set

Digit set	Type	Numbers formed	Count
$\{1, 4, 9\}$	all different	149, 194, 419, 491, 914, 941	6
$\{2, 3, 6\}$	all different	236, 263, 326, 362, 623, 632	6
$\{1, 6, 6\}$	two equal	166, 616, 661	3
$\{2, 2, 9\}$	two equal	229, 292, 922	3
$\{3, 3, 4\}$	two equal	334, 343, 433	3
		Total	**21**

Step 3. Add up
$6 + 6 + 3 + 3 + 3 = 21$ numbers.

Common mistake. Two common mistakes: (1) forgetting the sets containing the digit 1 (like $\{1, 4, 9\}$, $\{1, 6, 6\}$) because you only think of splitting 36 into three numbers "bigger than 1"; (2) assuming a set with two equal digits also gives 6 numbers – this double counts (for example, 334 would be counted twice).

Check: The table lists all 21 numbers, including 334 and 419 from the question. A computer check (going through every number from 100 to 999) also gives exactly 21.
Choose option **(A)**.

Answer: (A) 21

Problem 22

An *arithmetic sequence* is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, $2, 4, 6, 8$ and $1, 4, 7, 10$ are arithmetic sequences.
In the grid shown, the numbers in each row (read from left to right) must form an arithmetic sequence, and the numbers in each column (read from top to bottom) must form an arithmetic sequence. The value of x is

1			
4			25
7			x
10		36	

(A) 37 (B) 28 (C) 36 (D) 43.75 (E) 46

Detailed solution

Understanding the problem. The grid has 4 rows and 4 columns. Every row and every column is an arithmetic sequence. We need x in row 3, column 4. Idea: use the rows that already have two known numbers to find more numbers, then use column 4 to find x.

Recall. An arithmetic sequence with 4 terms can be written $a,\ a+d,\ a+2d,\ a+3d$ (d is the constant difference). From the first to the last term there are 3 steps, so $d = (\text{last} - \text{first}) \div 3$. If we know the first and the third terms, then $d = (\text{third} - \text{first}) \div 2$.

Step 1. Row 2: 4, _, _, 25
From 4 to 25 there are 3 steps: $d = (25 - 4) \div 3 = 21 \div 3 = 7$. Row 2 is 4, 11, 18, 25.

Step 2. Row 4: 10, _, 36, _
From the first term 10 to the third term 36 there are 2 steps: $d = (36 - 10) \div 2 = 26 \div 2 = 13$. Row 4 is 10, 23, 36, 49. So the last cell of row 4 is 49.

Step 3. Column 4: _, 25, x, 49
In column 4, x lies between 25 and 49, the same distance from each. So x is their mean:

$$x = \frac{25 + 49}{2} = \frac{74}{2} = 37.$$

(The difference in column 4 is $37 - 25 = 12$, so the top of the column is $25 - 12 = 13$.)

Step 4. Fill in the whole grid to check

1	5	9	13	+4
4	11	18	25	+7
7	17	27	37	+10
10	23	36	49	+13

Column 1: $1, 4, 7, 10$ (step 3); column 2: $5, 11, 17, 23$ (step 6); column 3: $9, 18, 27, 36$ (step 9); column 4: $13, 25, 37, 49$ (step 12). Every row and every column is arithmetic.

Common mistake. You cannot find x by looking at row 3 first: row 3 has only one known number (7), not enough to find its difference. You must go "around" through row 2 and row 4 to reach column 4.

Check: Row 3: $7, 17, 27, 37$ has constant difference 10; column 4: $25, 37, 49$ has constant difference 12. Correct.
$x = 37$, which is option **(A)**.

Answer: (A) 37

Problem 23

In the six-digit number $\overline{1ABCDE}$, each letter represents a digit. Given that $\overline{1ABCDE} \times 3 = \overline{ABCDE1}$, the value of $A + B + C + D + E$ is

(A) 29 (B) 26 (C) 22 (D) 30 (E) 28

Detailed solution

Understanding the problem. When $\overline{1ABCDE}$ is multiplied by 3, the leading digit 1 "moves" to the end, and the block $ABCDE$ shifts one place to the left. We must find the digits A, B, C, D, E and add them. There are two methods; both are well worth learning.

Recall. Place value: writing a digit c to the right of a number N gives $10 \cdot N + c$; writing the digit 1 to the left of a five-digit number N gives $100\,000 + N$.

Method 1: One unknown for the whole block.

Step 1. Let $N = \overline{ABCDE}$ (a five-digit number)
Then $\overline{1ABCDE} = 100\,000 + N$ and $\overline{ABCDE1} = 10N + 1$.

Step 2. Write the equation and find N

$$3 \cdot (100\,000 + N) = 10N + 1 \implies 300\,000 + 3N = 10N + 1.$$

Subtract $3N$ and 1 from both sides: $299\,999 = 7N$, so $N = 299\,999 \div 7 = 42\,857$.

Step 3. Read off the digits
$A = 4,\ B = 2,\ C = 8,\ D = 5,\ E = 7$. Sum: $4 + 2 + 8 + 5 + 7 = 26$.

Method 2: Multiply column by column, from right to left.

$$\begin{array}{r} 1\ A\ B\ C\ D\ E \\ \times \qquad\qquad 3 \\ \hline A\ B\ C\ D\ E\ 1 \end{array}$$

- Units: $3 \cdot E$ ends in $1 \Rightarrow E = 7$ ($3 \cdot 7 = 21$, write 1 carry 2).
- Tens: $3 \cdot D + 2$ ends in $E = 7 \Rightarrow 3D$ ends in $5 \Rightarrow D = 5$ ($15 + 2 = 17$, write 7 carry 1).

- Hundreds: $3 \cdot C + 1$ ends in $D = 5 \Rightarrow 3C$ ends in $4 \Rightarrow C = 8$ ($24 + 1 = 25$, write 5 carry 2).
- Thousands: $3 \cdot B + 2$ ends in $C = 8 \Rightarrow 3B$ ends in $6 \Rightarrow B = 2$ ($6 + 2 = 8$, carry 0).
- Ten thousands: $3 \cdot A + 0$ ends in $B = 2 \Rightarrow A = 4$ (12, write 2 carry 1).
- Hundred thousands: $3 \cdot 1 + 1 = 4$, exactly $A = 4$. The multiplication matches completely.

(At each step only one digit works, because $3 \cdot 0, 3 \cdot 1, \ldots, 3 \cdot 9$ all have different last digits.)

ThinkAlgo Tip. 142 857 is a famous number: $142\,857 \times 2 = 285\,714$, $\times 3 = 428\,571$, $\times 4 = 571\,428$, $\times 5 = 714\,285$, $\times 6 = 857\,142$ – always the digits $1, 4, 2, 8, 5, 7$ "rotated". It is linked to $\frac{1}{7} = 0.\overline{142857}$.

Check: $142\,857 \times 3 = 428\,571$. This is indeed $\overline{ABCDE1}$ with $ABCDE = 42857$.
$A + B + C + D + E = 26$, which is option **(B)**.

Answer: (B) 26

Problem 24

On a coordinate grid, a dot starts at the point $(20, 19)$. On its first move it goes 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On its second move it goes 2 units in a direction perpendicular to the first move (either way). Each move after that is 1 unit longer than the previous one (3 units, 4 units, 5 units,...) and must be perpendicular to the direction of the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Detailed solution

Understanding the problem. "Perpendicular to the previous move" means the moves *alternate*: horizontal, vertical, horizontal, vertical,... (or vertical, horizontal, vertical,...). The move lengths are $1, 2, 3, \ldots, 10$. We track separately the change in the x-coordinate (from horizontal moves) and in the y-coordinate (from vertical moves).

Recall. For the point (x, y): x is the horizontal coordinate (moving right increases x, moving left decreases it), y is the vertical coordinate (moving up increases y, moving down decreases it). A sum of an odd number of odd numbers is odd; a sum (or difference) of even numbers is always even.

Step 1. Split the moves into two groups
Moves $1, 3, 5, 7, 9$ are all in one direction (horizontal or vertical), with lengths $1, 3, 5, 7, 9$ (all odd). Moves $2, 4, 6, 8, 10$ are in the other direction, with lengths $2, 4, 6, 8, 10$ (all even).

- If the first move is *horizontal*: x changes by $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$ and y changes by $\pm 2 \pm 4 \pm 6 \pm 8 \pm 10$.
- If the first move is *vertical*: the other way round, x changes by $\pm 2 \pm 4 \pm 6 \pm 8 \pm 10$ and y by $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$.

(The sign + or − depends on which way the dot goes.)

Step 2. Odd and even

$\pm1\pm3\pm5\pm7\pm9$ combines 5 odd numbers, so it is always *odd*, and at most 25. And $\pm2\pm4\pm6\pm8\pm10$ is always *even*, at most 30. So of the two changes (in x and in y), exactly one is odd (from the odd group) and the other is even (from the even group).

Why is it true? Start with the "all plus" total $1+3+5+7+9=25$. Switching the sign of a number k from + to − lowers the total by $2k$. So every possible value is $25-2s$, where s is the sum of the numbers whose sign was switched. Likewise, the even group gives the values $30-2s$. □

Step 3. Compute the changes for each option

Subtract $(20, 19)$ from the final point:

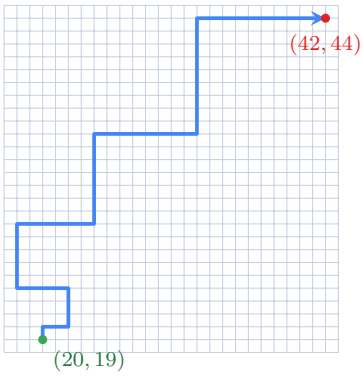
Option	Change in x	Change in y	Can it be made?
(A) $(27, 33)$	7	14	$1+3+5+7-9=7$; $2+4+6-8+10=14$
(B) $(30, 40)$	10	21	we would need $\pm1\pm3\pm5\pm7\pm9=21$ …
(C) $(21, 21)$	1	2	$1-3+5+7-9=1$; $2-4+6+8-10=2$
(D) $(42, 44)$	22	25	$2-4+6+8+10=22$; $1+3+5+7+9=25$
(E) $(37, 37)$	17	18	$-1-3+5+7+9=17$; $2+4-6+8+10=18$

Step 4. Why (B) is impossible

In (B) the odd change is 21, so we would need $\pm1\pm3\pm5\pm7\pm9=21$, i.e. $25-2s=21$, so $s=2$: we would have to switch the signs of some odd numbers adding up to 2. But a single odd number is never 2 (it can only be $1, 3, 5, \ldots$), and two different odd numbers of the group add up to at least $1+3=4$. There is no way! So the dot cannot stop at $(30, 40)$.

Step 5. An example path (option D)

First move up: up 1, right 2, up 3, left 4, up 5, right 6, up 7, right 8, up 9, right 10. The dot goes from $(20, 19)$ to $(42, 44)$.



Common mistake. Checking only "is the point too far away?" is not enough: $(30, 40)$ is no farther than $(42, 44)$. What decides it is odd/even and which totals can actually be made.

Check: Every equation in the table can be recomputed, e.g. $-1-3+5+7+9=17$. A computer search over all $2 \cdot 2^{10} = 2048$ ways of moving also confirms that only $(30, 40)$ cannot be reached.
Choose option **(B)**.

Answer: (B) $(30, 40)$

Problem 25

In the "Coin Game", you toss three coins at the same time. You win only if the 3 coins all show heads or all show tails. If you play the game once only, what is the probability of winning?

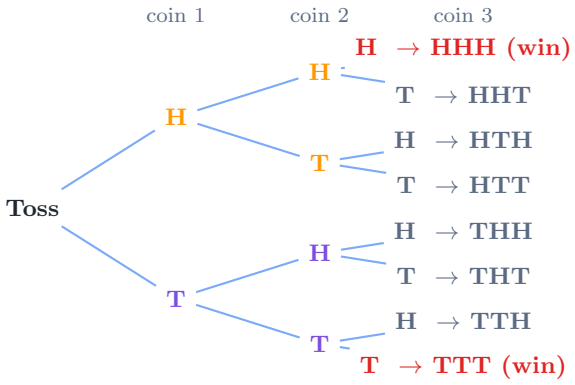
(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{2}{27}$ (D) $\frac{2}{3}$ (E) $\frac{1}{3}$

Detailed solution

Understanding the problem. Each coin has 2 faces: heads (H) or tails (T), equally likely. You win with HHH or TTT. We list all possible outcomes and count.

Recall. When all outcomes are equally likely: the *probability* of an event = (number of favourable outcomes) ÷ (total number of possible outcomes).

Step 1. List all outcomes with a tree diagram



The first coin has 2 possibilities, for each of them the second coin has 2, and then the third coin has 2: altogether $2 \cdot 2 \cdot 2 = 8$ equally likely outcomes.

Step 2. Count the favourable outcomes
Only 2 outcomes win: HHH and TTT.

Step 3. Compute the probability

$$P(\text{win}) = \frac{2}{8} = \frac{1}{4}.$$

Common mistake. A common *wrong* argument: "The number of heads can be $0, 1, 2, 3$ – four cases, and we win in 2 of them (0 and 3), so the probability is $\frac{2}{4}$." This is wrong because these four cases are *not* equally likely: "1 head" happens in 3 ways (HTT, THT, TTH), while "3 heads" happens in only 1 way.

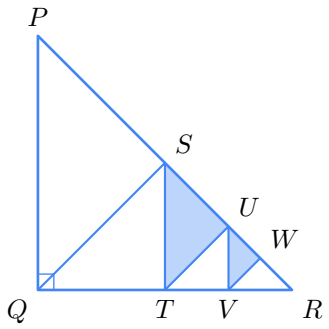
ThinkAlgo Tip. A quick way: the first coin can show anything. To win, the second coin must match the first (probability $\frac{1}{2}$) and the third must match too (probability $\frac{1}{2}$): $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Check: The 8 outcomes in the tree are all different and cover everything; exactly 2 of them win. $\frac{2}{8} = \frac{1}{4}$.
Choose option **(B)**.

Answer: (B) $\frac{1}{4}$

Problem 26

In the right-angled triangle PQR, the right angle is at Q and $PQ = QR$. The segments QS, TU and VW are perpendicular to PR, and the segments ST and UV are perpendicular to QR, as shown. What fraction of the area of triangle PQR is shaded?



(A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{5}{16}$ (D) $\frac{5}{32}$ (E) $\frac{7}{32}$

Detailed solution

Understanding the problem. Triangle PQR is a *right isosceles* triangle with the right angle at Q. The picture is built by repeatedly "dropping a perpendicular". The shaded part consists of two triangles: STU and UVW. Key idea: *dropping a perpendicular from the right-angle vertex of a right isosceles triangle cuts it into two equal halves, and each half is again a right isosceles triangle.*

Recall. In an isosceles triangle, the altitude from the apex (to the base) is also the median, and it divides the triangle into two congruent triangles. A right isosceles triangle has two acute angles of 45°.

Why is it true? Take triangle XYZ, right isosceles at X ($XY = XZ$), and draw $XH \perp YZ$. The right triangles XHY and XHZ have equal hypotenuses $XY = XZ$ and the common leg XH, so they are congruent (hypotenuse–leg). So each has half the area of XYZ. Moreover $\widehat{Y} = 45°$ and $\widehat{XHY} = 90°$, so $\widehat{HXY} = 45°$: triangle XHY is right isosceles at H. The same holds for XHZ. □

Step 1. Keep halving

Let the area of triangle PQR be 1 (the whole triangle).

- $QS \perp PR$ cuts PQR into two halves: triangle QSR has area $\frac{1}{2}$ and is right isosceles at S.
- $ST \perp QR$ cuts QSR into two halves: triangle STR has area $\frac{1}{4}$ and is right isosceles at T.
- $TU \perp SR$ cuts STR into two halves: triangle STU **(shaded)** has area $\frac{1}{8}$; triangle TUR has area $\frac{1}{8}$ and is right isosceles at U.
- $UV \perp TR$ cuts TUR into two halves: triangle UVR has area $\frac{1}{16}$ and is right isosceles at V.
- $VW \perp UR$ cuts UVR into two halves: triangle UVW **(shaded)** has area $\frac{1}{32}$.

Step 2. Add the shaded areas

$$\frac{1}{8} + \frac{1}{32} = \frac{4}{32} + \frac{1}{32} = \frac{5}{32}.$$

Step 3. Check with a square grid

Put Q at the origin with legs of 8 units: $Q(0,0)$, $R(8,0)$, $P(0,8)$. Then $S(4,4)$, $T(4,0)$, $U(6,2)$, $V(6,0)$, $W(7,1)$ (as in the figure). Triangle PQR has area $\frac{8 \cdot 8}{2} = 32$ squares. Triangle STU has side $ST = 4$ (vertical) and U is 2 units away from ST, so its area is $\frac{4 \cdot 2}{2} = 4$ squares. Triangle UVW has $UV = 2$ and W is 1 unit from UV, area $\frac{2 \cdot 1}{2} = 1$ square. Shaded: $4 + 1 = 5$ squares out of 32, i.e. $\frac{5}{32}$.

Recall. Area of a triangle $= \frac{\text{base} \times \text{height}}{2}$.

Common mistake. Many students pick $\frac{3}{16}$ or $\frac{3}{8}$ by "estimating from the picture". Always compute with the chain of halvings: each smaller triangle is half of the triangle containing it.

Check: Both methods (halving and counting squares) give $\frac{5}{32}$.
Choose option **(D)**.

Answer: (D) $\frac{5}{32}$

Problem 27

Mai went to a second-hand market where, every 15 minutes, the price of every item was reduced by 10% of its *current* price. At 9:00 a.m., the price of a rug was 100 000 dong. At 9:15 a.m., the price was reduced to 90 000 dong. As soon as the price of the rug fell *below* 80 000 dong, Mai bought it. At what time did Mai buy the rug?

(A) 9:45 a.m. (B) 9:15 a.m. (C) 9:30 a.m.

(D) 10:15 a.m. (E) 10:00 a.m.

Detailed solution

Understanding the problem. Each reduction is 10% of the *current price* (not of the original 100 000 dong). We compute the price after each 15-minute mark until it is less than 80 000 dong.

Recall. Reducing a number by 10% means taking away $\frac{10}{100} = \frac{1}{10}$ of it; what remains is 90% of it. So: new price = old price $\times 0.9$ (or old price $-$ old price $\div 10$).

Step 1. Make a table of prices over time

Time	Price before	Reduction (10%)	New price
9:00 a.m.	–	–	100 000
9:15 a.m.	100 000	10 000	90 000
9:30 a.m.	90 000	9 000	81 000
9:45 a.m.	81 000	8 100	**72 900**

Step 2. Find the first time the price is below 80 000 dong
At 9:30 a.m. the price is 81 000 dong – still *more* than 80 000 dong, so Mai does not buy yet. At 9:45 a.m. the price is 72 900 dong $<$ 80 000 dong: this is the first time it falls below 80 000 dong. Mai buys it at 9:45 a.m.

Common mistake. (1) The reduction is 10% *of the current price*, so the amount taken off gets smaller each time (10 000, then 9 000, then 8 100...); it is not 10 000 dong every time. (2) At 9:30 a.m. the price 81 000 dong is "very close" to 80 000 but not yet *below* 80 000, so it is not the moment to buy.

Check: $100\,000 \times 0.9 \times 0.9 \times 0.9 = 72\,900$ (after 3 reductions, i.e. $3 \times 15 = 45$ minutes after 9:00); after 2 reductions: $100\,000 \times 0.81 = 81\,000 > 80\,000$.
Choose option **(A)**.

Answer: (A) 9:45 a.m.

Problem 28

The number 2018 is used to create six-digit positive integers: each of them must contain the digits $2, 0, 1, 8$ *together and in this order*. For example, 720 186 is allowed, but 209 318 and 210 893 are not. How many of these six-digit integers are divisible by 9?

(A) 28 (B) 27 (C) 31 (D) 34 (E) 22

Detailed solution

Understanding the problem. The six-digit number contains the block "2018" plus two more digits (call them a and b). We must count those divisible by 9.

> **Recall. Divisibility by 9:** a number is divisible by 9 exactly when the sum of its digits is divisible by 9. The first digit of a six-digit number cannot be 0.

Step 1. Condition on the digit sum

The digits of the block add up to $2+0+1+8=11$. The whole number has digit sum $11+a+b$. Since a, b are digits, $0 \le a+b \le 18$, so $11 \le 11+a+b \le 29$. The multiples of 9 in this range are 18 and 27. Therefore

$$a+b=7 \qquad \text{or} \qquad a+b=16.$$

The only pairs with sum 16 are $(7,9),\ (8,8),\ (9,7)$.

Step 2. Three possible positions of the block 2018

Form	Extra condition	$a+b=7$	$a+b=16$	Total
$\overline{2018ab}$	none (a may be 0)	8 pairs: $a=0,1,\ldots,7$	3	11
$\overline{a2018b}$	$a \neq 0$	7 pairs: $a=1,\ldots,7$	3	10
$\overline{ab2018}$	$a \neq 0$	7 pairs: $a=1,\ldots,7$	3	10
			Total	**31**

About the column "$a+b=7$": the pairs are $(0,7),(1,6),(2,5),(3,4),(4,3),(5,2),(6,1),(7,0)$ – that is 8 pairs. When a is the first digit of the number, we drop $(0,7)$, leaving 7 pairs. The pairs with sum 16 all have $a \ge 7$, so none is dropped.

Step 3. Is any number counted twice?

A number would be counted twice only if the block 2018 appeared in two different places in the same six-digit number. Two 4-digit blocks inside 6 digits must overlap, and the block 2018 cannot overlap itself (for example, shifting it by 1 place would force the digit 0 to coincide with the digit 2...). So no number is counted twice.

> **Common mistake.** It is easy to forget one of the two cases: many students only consider $a+b=7$ (getting $8+7+7=22$, which is exactly the distractor (E)) and forget $a+b=16$. It is also easy to forget that the first digit cannot be 0 (that would give 34).

Check: A few examples: 201 870 has digit sum 18; 720 186 has digit sum 24 – not divisible by 9 (correct, since $7+6=13$ is neither 7 nor 16); 920 187 has digit sum 27, divisible by 9. A computer search of all numbers from 100 000 to 999 999 also gives exactly 31.

Choose option **(C)**.

Answer: (C) 31

Problem 29

The number N is the product of all positive odd integers from 1 to 99 that do *not* end in the digit 5. That is,

$$N = 1 \times 3 \times 7 \times 9 \times 11 \times 13 \times 17 \times 19 \times \cdots \times 91 \times 93 \times 97 \times 99.$$

The units digit of N is

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Detailed solution

Understanding the problem. N is a gigantic number that we cannot compute directly. But we only need its *last digit*, and the last digit of a product depends only on the last digits of the factors.

Recall. The units digit of a product equals the units digit of the product of the units digits of the factors. Example: $13 \times 17 = 221$ ends in 1, just like $3 \times 7 = 21$.

Step 1. Which units digits do the factors have?
Odd numbers that do not end in 5 can only end in $1, 3, 7, 9$. Split the factors into 10 groups, one for each ten:

$$(1 \cdot 3 \cdot 7 \cdot 9),\ (11 \cdot 13 \cdot 17 \cdot 19),\ (21 \cdot 23 \cdot 27 \cdot 29),\ \ldots,\ (91 \cdot 93 \cdot 97 \cdot 99).$$

Each group has 4 factors ending in $1, 3, 7, 9$; there are 10 groups (the tens 0 to 9), 40 factors in all.

Step 2. Units digit of each group
$1 \cdot 3 \cdot 7 \cdot 9 = 189$, which ends in 9. So the product of each group ends in 9.

Step 3. Multiply the 10 groups
The units digit of N equals the units digit of $\underbrace{9 \cdot 9 \cdots 9}_{10 \text{ factors}} = 9^{10}$. Now $9 \cdot 9 = 81$ ends in 1. Pair up the 10 nines into 5 pairs, each ending in 1, so 9^{10} ends in $1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$.

ThinkAlgo Tip. Powers of 9 have alternating last digits: $9^1 \to 9$, $9^2 \to 1$, $9^3 \to 9$, $9^4 \to 1, \ldots$: odd exponents end in 9, even exponents end in 1.

Common mistake. Do not think "a product of odd numbers ends in 9", and do not choose 5 by reflex: the numbers ending in 5 have been left out. Also count the groups correctly: from 1 to 99 there are 10 groups (the first one is $1, 3, 7, 9$).

Check: Step by step: after group 1 the last digit is 9, after group 2 it is $9 \cdot 9 \to 1$, after group 3 it is $9, \ldots$, after group 10 (an even number) it is 1. A computer computing N exactly also gives last digit 1.
Choose option **(A)**.

Answer: (A) 1

Problem 30

Five students wrote a quiz with a maximum score of 50. The scores of four of the students were 42, 43, 46 and 49. The score of the fifth student was N. The mean of the five students' scores was the same as the median of the five students' scores. The number of possible values of N is

(A) 3 (B) 4 (C) 1 (D) 0 (E) 2

Detailed solution

Understanding the problem. N is a score from 0 to 50. The mean depends on N; the median also depends on where N falls when the five scores are put in order. We consider each case for the position of N.

> **Recall.** The **median** of five numbers is the *middle* one (the third) when they are arranged from smallest to largest. The **mean** = sum ÷ how many numbers.

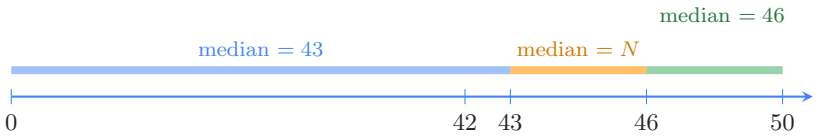
Step 1. Write the mean in terms of N

The four known scores add to $42 + 43 + 46 + 49 = 180$. The mean of the five scores is $\dfrac{180 + N}{5}$.

Step 2. What is the median? – Three cases

Arrange $42, 43, 46, 49$ together with N in increasing order:

- If $N \leq 43$: the order is $\ldots, 43, 46, 49$ with N in one of the first two places; the third number is 43. Median = 43.
- If $43 \leq N \leq 46$: the order is $42, 43, N, 46, 49$. Median = N.
- If $N \geq 46$: the order is $42, 43, 46, \ldots$ with N in one of the last two places. Median = 46.



Step 3. Solve each case

- **Case 1** (median 43): $\dfrac{180 + N}{5} = 43 \Rightarrow 180 + N = 215 \Rightarrow N = 35$. Check: $35 \leq 43$ – it is in the case being considered. **Accept.**
- **Case 2** (median N): $\dfrac{180 + N}{5} = N \Rightarrow 180 + N = 5N \Rightarrow 4N = 180 \Rightarrow N = 45$. Check: $43 \leq 45 \leq 46$. **Accept.**
- **Case 3** (median 46): $\dfrac{180 + N}{5} = 46 \Rightarrow 180 + N = 230 \Rightarrow N = 50$. Check: $50 \geq 46$, and 50 does not exceed the maximum score. **Accept.**

> **Common mistake.** After solving for N in each case, you *must check* that N really belongs to that case (and lies between 0 and 50). In this problem all three values are valid, but in many other problems some values get rejected. Do not stop after finding one value!

Check: $N = 35$: scores $35, 42, 43, 46, 49$, sum 215, mean 43, median 43. $N = 45$: scores $42, 43, 45, 46, 49$, sum 225, mean 45, median 45. $N = 50$: scores $42, 43, 46, 49, 50$, sum 230, mean 46, median 46.
There are 3 values of N, which is option **(A)**.

Answer: **(A)** 3

Extension and challenge

Problem 31

Consider the numbers: -1.5; 0.25; -0.75; 1.2; -2.

a) Arrange the numbers in increasing order.

b) Which numbers lie between -1 and 1 on the number line?

c) Find the sum of the largest and the smallest number.

Detailed solution

Understanding the problem. This problem combines two skills from Problems 1 and 3: comparing decimals and comparing negative numbers.

> **Recall.** Negative numbers are always less than positive numbers. Of two negative numbers, the one farther from 0 is smaller. For two positive numbers, compare the whole parts, then the decimal places one by one.

Step 1. Split into two groups: negative and positive

Negative: -1.5; -0.75; -2. Positive: 0.25; 1.2.

Step 2. Order the negative numbers

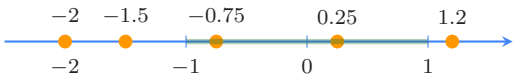
Distance from 0: -2 is 2 away; -1.5 is 1.5 away; -0.75 is 0.75 away. The one farthest from 0 is the smallest:

$$-2 < -1.5 < -0.75.$$

Step 3. Order the positive numbers, then put everything together

$0.25 < 1.2$ (whole parts $0 < 1$). Negative numbers first, positive numbers after:

$$-2 < -1.5 < -0.75 < 0.25 < 1.2.$$



Step 4. The numbers between -1 and 1

On the number line, the green segment from -1 to 1 contains -0.75 and 0.25. (-1.5 is to the left of -1; 1.2 is to the right of 1.)

Step 5. Sum of the largest and the smallest

The largest is 1.2, the smallest is -2:

$$1.2 + (-2) = -(2 - 1.2) = -0.8.$$

> **Recall.** Adding two numbers with opposite signs: subtract the smaller size from the larger (ignoring signs), then use the sign of the number farther from 0.

> **Common mistake.** Many students put -0.75 before -1.5 because "$0.75 < 1.5$". For negative numbers it is reversed: $-1.5 < -0.75$.

Check: $-0.8 - 1.2 = -2$. Correct.

Answer: **a)** $-2 < -1.5 < -0.75 < 0.25 < 1.2$; **b)** -0.75 **and** 0.25; **c)** -0.8

Problem 32

Consider the expression $2 + 3 \times 4 - 1$.

a) Evaluate the expression.

b) Insert *one* pair of brackets ( ) into the expression (without changing the order of the numbers or the signs) to get the largest possible value. What is that largest value?

c) Same question as b), but for the smallest value.

Detailed solution

Understanding the problem. This extends Problems 4 and 7: the position of the brackets changes the order of the operations, and so changes the result.

Recall. Brackets first; then multiplication and division; finally addition and subtraction (from left to right).

Step 1. Value without brackets

$2 + 3 \times 4 - 1 = 2 + 12 - 1 = 13$.

Step 2. List systematically every way to place one pair of brackets

A pair of brackets must enclose at least two neighbouring numbers. Call the four numbers, in order, number $1, 2, 3, 4$ (2; 3; 4; 1). The brackets can start at number i and end at number j with $i < j$: 6 ways.

Brackets around	Expression	Value
numbers 1 – 2	$(2 + 3) \times 4 - 1 = 5 \times 4 - 1$	**19**
numbers 1 – 3	$(2 + 3 \times 4) - 1 = 14 - 1$	13
numbers 1 – 4	$(2 + 3 \times 4 - 1)$	13
numbers 2 – 3	$2 + (3 \times 4) - 1$	13
numbers 2 – 4	$2 + (3 \times 4 - 1) = 2 + 11$	13
numbers 3 – 4	$2 + 3 \times (4 - 1) = 2 + 9$	**11**

Step 3. Read off the results

The largest is 19, with $(2 + 3) \times 4 - 1$: the sum $2 + 3$ is "blown up" 4 times.
The smallest is 11, with $2 + 3 \times (4 - 1)$: 3 is multiplied by a smaller number (3 instead of 4).

ThinkAlgo Tip. For a large result, let the multiplication act on a large sum; for a small result, let it act on a difference. But to be sure, the best way is still to list every case, as in the table.

Check: $(2 + 3) \times 4 - 1 = 20 - 1 = 19$ and $2 + 3 \times 3 = 11$.

Answer: **a)** 13; **b)** $(2 + 3) \times 4 - 1 = 19$; **c)** $2 + 3 \times (4 - 1) = 11$

Problem 33

An's storybook has 360 pages. On the first day An reads $\frac{1}{4}$ of the book. On the second day An reads $\frac{1}{3}$ *of the remaining pages*. After two days, how many pages has An not read yet, and what fraction of the book is that?

Detailed solution

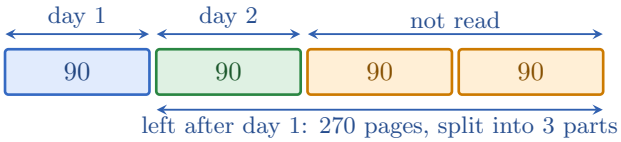
Understanding the problem. Careful: the fraction $\frac{1}{3}$ on the second day is taken of the *remaining pages*, not of the whole book. This is a "two-step" version of Problem 9.

Recall. $\frac{m}{n}$ of a equals $a \div n \times m$.

Step 1. First day
An reads $360 \div 4 = 90$ (pages). Remaining: $360 - 90 = 270$ (pages).

Step 2. Second day (taken of 270 pages)
An reads $270 \div 3 = 90$ (pages). Remaining: $270 - 90 = 180$ (pages).

Step 3. Fraction of the book left
$\frac{180}{360} = \frac{1}{2}$.



Common mistake. A common mistake: adding $\frac{1}{4} + \frac{1}{3} = \frac{7}{12}$ and saying $\frac{5}{12}$ of the book (150 pages) is left. Wrong, because $\frac{1}{3}$ is of the *remaining part* (270 pages), not of the whole book (360 pages).

Step 4. Another way: multiply the "remaining" fractions
After day 1, $\frac{3}{4}$ of the book remains; after day 2, $\frac{2}{3}$ of that remains, i.e. $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}$ of the book, which is $360 \div 2 = 180$ pages.

Check: $90 + 90 + 180 = 360$. Correct.

Answer: 180 pages, which is $\frac{1}{2}$ of the book

Problem 34

The first test has 40 questions, and Bình answers 32 of them correctly.

a) What percentage of the questions in the first test did Bình answer correctly?

b) The second test has 60 questions. At least how many questions must Bình answer correctly in

the second test so that, over both tests together, his correct answers make up at least 85% of all the questions?

Detailed solution

Understanding the problem. Part a) asks for a percentage (the reverse of Problem 5). Part b) asks for a percentage of a total.

> **Recall.** a as a percentage of b is $\frac{a}{b} \times 100\%$. $p\%$ of a is $a \times \frac{p}{100}$.

Step 1. Percentage in the first test

$\frac{32}{40} \times 100\% = 0.8 \times 100\% = 80\%$.

Step 2. Total number of questions in both tests

$40 + 60 = 100$ (questions).

Step 3. Correct answers needed overall

85% of 100 questions is $100 \times \frac{85}{100} = 85$ (questions).

Step 4. Correct answers needed in the second test

Bình already has 32 correct answers, so he needs at least $85 - 32 = 53$ (questions) in the second test.

> **Common mistake.** Do not take "the average of the two percentages": solving $(80\% + x) \div 2 = 85\%$ gives $x = 90\%$ of 60 questions, i.e. 54 questions – wrong! The tests have different numbers of questions, so we must work with the *total number of questions*.

Check: With 53 correct: $(32+53) \div 100 = 85\%$ (just enough). With 52: $84 \div 100 = 84\% < 85\%$ (not enough).

Answer: a) 80%; b) at least 53 questions

Problem 35

A shop sells cashew nuts in packs: a 3 kg pack costs 255 thousand dong, a 5 kg pack costs 410 thousand dong (no loose sales).

a) Which pack is cheaper per kilogram?

b) Ms Hằng needs to buy *exactly* 15 kg. How should she buy them to spend as little as possible? How much does it cost?

c) How much does it cost to buy *exactly* 13 kg?

Detailed solution

Understanding the problem. This extends Problem 6: compare unit prices, then find how to combine packs to get the exact mass.

> **Recall.** Unit price = total cost ÷ quantity.

Step 1. Unit price of each pack
3 kg pack: $255 \div 3 = 85$ (thousand dong/kg). 5 kg pack: $410 \div 5 = 82$ (thousand dong/kg).
So the 5 kg pack is cheaper ($82 < 85$).

Step 2. Ways to make exactly 15 kg
Let a be the number of 3 kg packs and b the number of 5 kg packs: $3a + 5b = 15$. Try $b = 0, 1, 2, 3$:

b	$15 - 5b$	a	Cost (thousand dong)
0	15	5	$5 \times 255 = 1275$
1	10	not divisible by 3	–
2	5	not divisible by 3	–
3	0	0	$3 \times 410 = \mathbf{1230}$

Buying 3 packs of 5 kg is cheapest: 1 230 thousand dong = 1 230 000 dong.

Step 3. Exactly 13 kg
$3a + 5b = 13$: $b = 0 \Rightarrow 13$ (not divisible by 3); $b = 1 \Rightarrow 8$ (not divisible by 3); $b = 2 \Rightarrow 3$, $a = 1$.
There is only one way: 1 pack of 3 kg and 2 packs of 5 kg.
Cost: $255 + 2 \times 410 = 255 + 820 = 1075$ (thousand dong) = 1 075 000 dong.

Common mistake. Part c) cannot be answered by computing 13×82, because the shop does not sell loose nuts: the packs must add up to exactly 13 kg.

Check: $3 + 5 + 5 = 13$ kg; $5 + 5 + 5 = 15$ kg.

Answer: a) the 5 kg pack (82 thousand dong/kg); b) 3 packs of 5 kg, 1 230 000 dong; c) 1 075 000 dong

Problem 36

The five members of a group weigh (in kg) 38; 41; 40; 37 and x. Their mean weight is 40 kg.

a) Find x. Is the fifth member heavier or lighter than the mean, and by how many kilograms?

b) Find the median weight of the group.

c) A sixth member joins the group, and the mean weight of all six becomes 41 kg. How much does the sixth member weigh?

Detailed solution

Understanding the problem. This extends Problem 14: use the sum and the deviations to find the missing number; add the median and an "extra member" question.

Recall. Sum = mean × how many numbers. The deviations from the mean add up to 0.

Step 1. Find x using deviations
Compared with 40: $38 \to -2$; $41 \to +1$; $40 \to 0$; $37 \to -3$. Sum: $-2 + 1 + 0 - 3 = -4$.
For the deviations to add up to 0, the fifth member must be $+4$: $x = 40 + 4 = 44$ (kg), 4 kg heavier than the mean.

Check with the sum: $5 \times 40 = 200$; $200 - (38 + 41 + 40 + 37) = 200 - 156 = 44$.

Step 2. Median

In increasing order: 37; 38; **40**; 41; 44. The middle (third) number is 40. The median is 40 kg.

Step 3. The sixth member

Total weight of six members: $6 \times 41 = 246$ (kg). The first five weigh 200 kg in total. The sixth weighs $246 - 200 = 46$ (kg).

ThinkAlgo Tip. Think in deviations: the mean rises by 1 kg for all 6 people, so 6 kg extra above the level of 40 are needed; the new member must weigh $40 + 6 = 46$ kg.

Common mistake. It is not "the sixth member weighs 41 kg" – then the mean would rise only a little: $241 \div 6 \approx 40.17$ kg.

Check: $(37 + 38 + 40 + 41 + 44 + 46) \div 6 = 246 \div 6 = 41$. Correct.

Answer: a) $x = 44$ **kg, 4 kg heavier than the mean; b)** 40 **kg; c)** 46 **kg**

Problem 37

A running track around the schoolyard is 200 m long. Mai runs 6 laps in 4 minutes. Tú runs at a constant 18 km/h.

a) Who is faster?

b) Keeping the same speeds, how many minutes does each of them need to run 3 km?

c) They start at the same time from the same point of the track, but run in *opposite directions*. After how many seconds do they meet for the first time?

d) One day Tú increases his speed to 21.6 km/h while Mai runs as before. They start at the same time from the same point, running in the *same direction*. How long is it until Tú is exactly one lap ahead of Mai?

Detailed solution

Understanding the problem. This extends Problem 12: find a speed from laps run, convert km/h to m/s, then study two runners on the same circular track.

Recall. $v = s \div t$, $t = s \div v$. 1 km/h $= \dfrac{5}{18}$ m/s, so to convert km/h to m/s multiply by 5 and divide by 18.

Step 1. Mai's speed

Distance: $6 \times 200 = 1200$ (m). Time: 4 minutes = 240 seconds. $v_{\text{Mai}} = 1200 \div 240 = 5$ (m/s).

Step 2. Tú's speed (in m/s)

$v_{\text{Tú}} = 18 \times \dfrac{5}{18} = 5$ (m/s). They run **equally fast**!

Common mistake. Comparing "5" with "18" directly is meaningless, because the units differ (m/s and km/h). Convert to the same unit first.

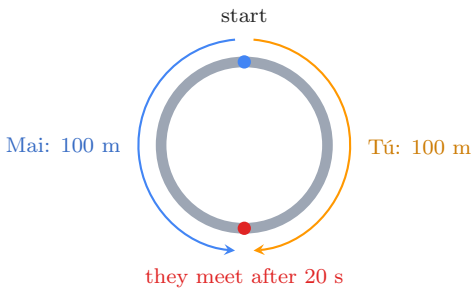
Step 3. Time to run 3 km

3 km = 3000 m. $t = 3000 \div 5 = 600$ (seconds) $= 600 \div 60 = 10$ (minutes). Both need 10 minutes.

Step 4. Opposite directions

Every second Mai runs 5 m and Tú runs 5 m the other way, so the "combined" distance they cover grows by $5 + 5 = 10$ m per second. They first meet when the *total* of their distances is exactly one lap, 200 m:

$$t = 200 \div 10 = 20 \text{ (seconds)}.$$



After 20 seconds each has run $5 \times 20 = 100$ m, exactly half a lap: they meet at the point opposite the start.

Step 5. Same direction, Tú faster

$v_{\text{Tú}} = 21.6 \times \frac{5}{18} = \frac{108}{18} = 6$ (m/s). Every second Tú runs $6 - 5 = 1$ m more than Mai. Tú is exactly one lap ahead when he has run 200 m more than Mai:

$$t = 200 \div 1 = 200 \text{ (seconds)} = 3 \text{ minutes } 20 \text{ seconds}.$$

Check: After 200 seconds: Tú has run $6 \times 200 = 1200$ m (6 laps), Mai $5 \times 200 = 1000$ m (5 laps). Tú is exactly 1 lap ahead.

Note for parents. You can act out parts c) and d) with two coins on a round plate: in opposite directions the "gap to close" shrinks by the *sum* of the speeds; in the same direction it shrinks by the *difference* of the speeds.

Answer: a) equally fast (5 m/s); b) 10 minutes; c) 20 seconds; d) 200 seconds (3 minutes 20 seconds)

Problem 38

a) Find the digits a, b such that the number $\overline{4a7b}$ is divisible by 2, 5 and 9.

b) How many four-digit numbers of the form $\overline{5a3b}$ are divisible by 9? List them.

Detailed solution

Understanding the problem. This extends Problem 10: use several divisibility rules at once and count systematically.

Recall. Divisible by 2 and 5 $\iff$ the last digit is 0. Divisible by 9 $\iff$ the digit sum is divisible by 9.

Why is it true? Why is "divisible by both 2 and 5" the same as "ends in 0"? Divisibility by 2 needs an even last digit; divisibility by 5 needs a last digit of 0 or 5. The only digit that is even and in $\{0;5\}$ is 0. □

Step 1. Part a) – find b
$\overline{4a7b}$ is divisible by 2 and 5, so $b = 0$.

Step 2. Part a) – find a
Digit sum: $4 + a + 7 + 0 = 11 + a$. Since $0 \le a \le 9$, we have $11 \le 11 + a \le 20$. The only multiple of 9 in this range is 18. So $11 + a = 18$, $a = 7$. The number is 4770.

Check: $4770 = 9 \times 530$, and it ends in 0, so it is divisible by 2 and 5.

Step 3. Part b) – the condition
The digit sum of $\overline{5a3b}$ is $5 + a + 3 + b = 8 + a + b$. Since $0 \le a + b \le 18$, we have $8 \le 8 + a + b \le 26$. The multiples of 9 in this range are 9 and 18. Hence $a + b = 1$ or $a + b = 10$.

Step 4. Part b) – count systematically
$a + b = 1$: $(a;b) = (0;1)$ or $(1;0)$ – 2 numbers.
$a + b = 10$: a runs from 1 to 9, $b = 10 - a$ (from 9 down to 1) – 9 numbers. ($a = 0$ would give $b = 10$, not a digit.)
In total: $2 + 9 = 11$ numbers:

$$5031,\ 5130,\ 5139,\ 5238,\ 5337,\ 5436,\ 5535,\ 5634,\ 5733,\ 5832,\ 5931.$$

Common mistake. Do not forget that a may be 0 (the first digit is 5), and do not miss the case where the digit sum is 9 (that is, $a + b = 1$).

Check: Each number above has digit sum 9 or 18, e.g. 5 436: $5 + 4 + 3 + 6 = 18$.

Answer: a) $a = 7$, $b = 0$ (the number 4770); b) 11 numbers

Problem 39 ★

In the addition below, M and N are two different non-zero digits. Find all pairs $(M;N)$ that work.

$$\begin{array}{r} M\ N \\ N\ M \\ +\ N\ N \\ \hline 1\ 8\ 7 \end{array}$$

Detailed solution

Understanding the problem. This extends Problem 13. This time, instead of trying column by column, we use place value to get one simple equation, and then find *all* solutions.

Recall. $\overline{MN} = 10M + N$, $\overline{NM} = 10N + M$, $\overline{NN} = 10N + N = 11N$.

Step 1. Write the sum using place value

$$\overline{MN} + \overline{NM} + \overline{NN} = (10M + N) + (10N + M) + 11N = 11M + 22N = 11 \times (M + 2N).$$

Step 2. Set up the equation
$11 \times (M + 2N) = 187$, so $M + 2N = 187 \div 11 = 17$.

Step 3. Find every pair of digits
$M = 17 - 2N$ must be a digit from 1 to 9, so $17 - 2N \leq 9$, i.e. $N \geq 4$; and $17 - 2N \geq 1$, i.e. $N \leq 8$.

N	4	5	6	7	8
$M = 17 - 2N$	9	7	5	3	1
Addition	$94 + 49 + 44$	$75 + 57 + 55$	$56 + 65 + 66$	$37 + 73 + 77$	$18 + 81 + 88$

All five pairs have $M \neq N$, and every sum equals 187.

ThinkAlgo Tip. When letters appear "symmetrically", as in $\overline{MN}$ and $\overline{NM}$, try place value: there is often a common factor (11 here) that makes the problem much simpler.

Common mistake. The problem asks for *all* pairs. Finding one pair (say $M = 9$, $N = 4$) and stopping is not enough.

Check: $94 + 49 + 44 = 187$; $75 + 57 + 55 = 187$; $56 + 65 + 66 = 187$; $37 + 73 + 77 = 187$; $18 + 81 + 88 = 187$.

Answer: $(M; N) \in \{(9; 4),\ (7; 5),\ (5; 6),\ (3; 7),\ (1; 8)\}$

Problem 40 ★★

A sequence is built by the Fibonacci rule: from the third number on, each number is the sum of the two numbers just before it.

a) The first number is 2 and the fifth number is 19. Find the second and the eighth numbers of the sequence.

b) Find the mean of the first five numbers of the sequence in part a).

c) In the familiar Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$, is the 2026th number even or odd? Explain.

Detailed solution

Understanding the problem. This extends Problem 15. In part a) the second number is unknown, so we call it b and express the following numbers in terms of b. In part c) we cannot write out 2026 numbers; we need a pattern of even and odd.

Step 1. Express the numbers in terms of b

Position	1	2	3	4	5
Number	2	b	$2+b$	$b+(2+b)=2+2b$	$(2+b)+(2+2b)=4+3b$

Step 2. Find b

$4+3b=19 \Rightarrow 3b=15 \Rightarrow b=5$. The second number is 5.

Step 3. Continue up to the eighth number

2, 5, 7, 12, 19, 31, 50, 81 $\quad(12+19=31;\ 19+31=50;\ 31+50=81)$. The eighth number is 81.

Step 4. Mean of the first five numbers

$(2+5+7+12+19)\div 5=45\div 5=9$.

Step 5. The even–odd pattern of the Fibonacci sequence

Recall. odd + odd = even; odd + even = odd; even + odd = odd.

Write O (odd) and E (even) for the numbers $1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$:

Position	1	2	3	4	5	6	7	8	9
Number	1	1	2	3	5	8	13	21	34
Even/odd	O	O	**E**	O	O	**E**	O	O	**E**

The pattern "O, O, E" repeats forever.

Why is it true? Suppose at some point we have two consecutive numbers O, O. Then the next number is O + O = E; the one after is O + E = O; the one after that is E + O = O. We again have two consecutive numbers O, O, so the cycle "O, O, E" starts again. The sequence begins with 1, 1 (O, O), so this pattern repeats all the way: the numbers in positions divisible by 3 are even, all the others are odd. □

Step 6. Look at position 2026

$2026 = 3\times 675+1$, i.e. 2026 leaves remainder 1 when divided by 3. This position corresponds to the first "O" in a cycle. So the 2026th number is **odd**.

Common mistake. You cannot conclude "2026 is even, so the 2026th number is even too". The parity of the *position* is different from the parity of the *number* in that position; for example, the 2nd number is 1 (odd).

Check: Part a): $2+5=7$, $5+7=12$, $7+12=19$ – indeed the fifth number is 19. Part c): test the pattern on small positions: position $4=3\times 1+1$ holds 3 (odd); position $7=3\times 2+1$ holds 13 (odd).

Answer: a) the second number is 5, the eighth is 81; b) 9; c) odd

Problem 41 *

Sums of consecutive numbers (extends Problem 16).

a) Find *all* the ways of writing 100 as a sum of at least two consecutive positive integers.

b) Prove that 64 cannot be written as a sum of at least two consecutive positive integers.

Detailed solution

Understanding the problem. A sum of consecutive positive integers is determined by two pieces of information: *how many numbers* (call it k, with $k \geq 2$) and *the first number* (call it a, with $a \geq 1$). We need a quick formula for such a sum, and then we use it to "narrow down" the possibilities.

Recall. Sum of an arithmetic sequence (Gauss's method): $\text{sum} = \dfrac{(\text{first} + \text{last}) \times \text{number of terms}}{2}$.

Why is it true? Write the sum twice, once forwards and once backwards, and add column by column: for example $S = 9 + 10 + \cdots + 16$ and $S = 16 + 15 + \cdots + 9$. Each column adds up to $9 + 16 = 25$ and there are 8 columns, so $2S = 25 \cdot 8$, $S = 100$. The same argument works for any arithmetic sequence. □

Step 1. Set up the key equation

With k consecutive numbers starting at a, the last number is $a + k - 1$. Let $m = (\text{first} + \text{last}) = 2a + k - 1$. By Gauss's formula: $\text{sum} = \dfrac{m \cdot k}{2}$. If the sum is 100, then

$$k \cdot m = 200.$$

Step 2. Two important observations about k and m

- $m - k = 2a - 1$ is odd, so k **and** m **have opposite parity** (one is even, the other odd).
- $m - k = 2a - 1 \geq 1$, so $m > k$.

Step 3. List the factorisations $200 = k \cdot m$ with $2 \leq k < m$

Since $k < m$ and $k \cdot m = 200$, we get $k \cdot k < 200$, so $k \leq 14$. The divisors of 200 from 2 to 14 are $2, 4, 5, 8, 10$.

k	$m = 200 \div k$	Opposite parity?	Result
2	100	no (both even)	rejected
4	50	no	rejected
5	40	yes	$2a + 4 = 40 \Rightarrow a = 18$
8	25	yes	$2a + 7 = 25 \Rightarrow a = 9$
10	20	no	rejected

Step 4. Write out the sums

$$100 = 18 + 19 + 20 + 21 + 22 \quad \text{and} \quad 100 = 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16.$$

There are exactly **two** ways.

ThinkAlgo Tip. Remember Problem 16: with 5 consecutive numbers (an odd number of terms), the middle number equals the mean $= 100 \div 5 = 20$, so the five numbers are $18, 19, 20, 21, 22$. The "middle number" idea gives a very quick check.

Step 5. Part b): why 64 is impossible
Suppose 64 is a sum of $k \geq 2$ consecutive positive integers. Arguing as above: $k \cdot m = 2 \cdot 64 = 128$ with $m > k \geq 2$ and k, m of opposite parity. So one of k, m is odd and divides 128. But $128 = 2^7$ has only *one* odd divisor, namely 1, while both k and m are ≥ 2. Contradiction! So 64 cannot be written in this way.

Why is it true? The same argument shows that *every power of* 2 $(2, 4, 8, 16, 32, 64, \dots)$ cannot be written as a sum of at least two consecutive positive integers. Conversely, every number with an odd divisor greater than 1 can be. □

Common mistake. Do not forget the word "positive": if 0 and negative numbers were allowed there would be extra "fake" ways, e.g. $(-17) + (-16) + \cdots + 21 + 22 = 100$ (the numbers from -17 to 17 cancel, leaving $18 + 19 + 20 + 21 + 22$). Here every number must be ≥ 1, so $a \geq 1$, which gives $m > k$.

Check: $18 + 19 + 20 + 21 + 22 = 100$; $9 + 10 + \cdots + 16 = \dfrac{(9 + 16) \cdot 8}{2} = 100$. Trying 64 by hand: 2 consecutive numbers have an odd sum; 3 consecutive numbers have a sum divisible by 3; ... no way gives 64.

Answer: a) Two ways: $18 + 19 + 20 + 21 + 22$ and $9 + 10 + \cdots + 16$. b) See the proof.

Problem 42 *

Common divisors and counting divisors (extends Problem 17).

a) How many positive integers divide both 720 and 1080?

b) Find the smallest positive integer with exactly 12 divisors.

Detailed solution

Understanding the problem. Listing the divisors of 720 and 1080 would be very long (each has dozens of divisors). We use two powerful tools: (1) the common divisors of two numbers are the divisors of their GCD; (2) the formula counting divisors from the prime factorisation.

Recall. Counting divisors. If $n = p^x \cdot q^y \cdot r^z$ (p, q, r different primes), then n has $(x + 1)(y + 1)(z + 1)$ divisors.

Why is it true? Each divisor of n has the form $p^i \cdot q^j \cdot r^l$ with $0 \le i \le x$, $0 \le j \le y$, $0 \le l \le z$. The exponent i has $x+1$ choices $(0, 1, \ldots, x)$, j has $y+1$ choices, l has $z+1$ choices. By the multiplication rule there are $(x+1)(y+1)(z+1)$ divisors, and different choices give different divisors (because prime factorisation is unique). □

Part a).

Step 1. Prime factorisation

$720 = 2^4 \cdot 3^2 \cdot 5$ (since $720 = 16 \cdot 45 = 16 \cdot 9 \cdot 5$); $1080 = 2^3 \cdot 3^3 \cdot 5$ (since $1080 = 8 \cdot 135 = 8 \cdot 27 \cdot 5$).

Step 2. Find the GCD

Take each common prime with the *smaller* exponent: $\mathrm{GCD}(720, 1080) = 2^3 \cdot 3^2 \cdot 5 = 8 \cdot 9 \cdot 5 = 360$. (Another check: $1080 - 720 = 360$ and $720 = 2 \cdot 360$, so the GCD is 360.)

Step 3. Count the divisors of 360

$360 = 2^3 \cdot 3^2 \cdot 5^1$ has $(3+1)(2+1)(1+1) = 4 \cdot 3 \cdot 2 = 24$ divisors. So 24 numbers divide both 720 and 1080.

Part b).

Step 4. The "shapes" of a number with exactly 12 **divisors**

The number of divisors is a product of factors (exponent +1). We write 12 as a product of numbers ≥ 2:

$$12 = 12, \qquad 12 = 6 \cdot 2, \qquad 12 = 4 \cdot 3, \qquad 12 = 3 \cdot 2 \cdot 2.$$

These correspond to four shapes: p^{11}, $p^5 q$, $p^3 q^2$, $p^2 qr$.

Step 5. The smallest number of each shape

To make the number small, use the smallest primes $2, 3, 5$, and give the *larger* exponent to the *smaller* prime:

Shape	Smallest number	Value
p^{11}	2^{11}	2048
$p^5 q$	$2^5 \cdot 3$	96
$p^3 q^2$	$2^3 \cdot 3^2$	72
$p^2 qr$	$2^2 \cdot 3 \cdot 5$	**60**

The smallest is 60.

Common mistake. Do not give the large exponent to the large prime: for example $2 \cdot 3^5 = 486$ is much bigger than $2^5 \cdot 3 = 96$. And do not forget the shape with *three* primes $(p^2 qr)$ – it is the one that gives the answer.

Check: The divisors of 60: $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$ – exactly 12. A computer check of the numbers from 1 to 59: none has 12 divisors.

Answer: a) 24 **numbers. b)** 60.

Problem 43 *

How does a fraction change? (extends Problems 18 and 19).

a) When the same whole number n is added to both the numerator and the denominator of $\frac{5}{8}$, the result is a fraction equal to $\frac{4}{5}$. Find n.

b) Let $\frac{a}{b}$ be a fraction with $0 < a < b$. Prove that for every positive integer n, $\frac{a+n}{b+n} > \frac{a}{b}$.

c) How many fractions equal to $\frac{3}{7}$ have a two-digit denominator?

Detailed solution

Understanding the problem. Parts a) and b) are both about "adding the same number to the numerator and the denominator". What does *not* change when we do that is the *difference* between the denominator and the numerator – this is the key.

Part a).

Step 1. Find the quantity that stays the same

For $\frac{5}{8}$, denominator − numerator is $8-5=3$. After adding n to both: $(8+n)-(5+n)=3$, still 3.

Step 2. Use a "parts" model as in Problem 19

The new fraction equals $\frac{4}{5}$, so the new numerator is 4 parts and the new denominator is 5 parts; the difference is 1 part. Since the difference is 3, 1 part = 3. New numerator $= 4 \cdot 3 = 12$, new denominator $= 5 \cdot 3 = 15$.

Step 3. Find n

$n = 12 - 5 = 7$ (and also $15 - 8 = 7$).

Check: $\frac{5+7}{8+7} = \frac{12}{15} = \frac{12 \div 3}{15 \div 3} = \frac{4}{5}$. Correct.

Part b).

Step 4. Compare "what is missing to reach 1"

Since $a < b$, both fractions are less than 1. What is missing to make 1 is

$$1 - \frac{a}{b} = \frac{b-a}{b}, \qquad 1 - \frac{a+n}{b+n} = \frac{(b+n)-(a+n)}{b+n} = \frac{b-a}{b+n}.$$

The fractions $\frac{b-a}{b}$ and $\frac{b-a}{b+n}$ have *the same positive numerator*, and the denominator $b+n > b$.

The fraction with the larger denominator is smaller, so $\frac{b-a}{b+n} < \frac{b-a}{b}$.

Step 5. Conclusion

The fraction $\frac{a+n}{b+n}$ is "missing less" to reach 1, so it is larger: $\frac{a+n}{b+n} > \frac{a}{b}$.

Recall. Two positive fractions with the same numerator: the one with the larger denominator is smaller (the same cake shared among more people gives smaller pieces).

Why is it true? Another way (cross-multiplying): with positive denominators, $\frac{a+n}{b+n} > \frac{a}{b}$ exactly when $(a+n)\cdot b > a\cdot(b+n)$, i.e. $ab+nb > ab+na$, i.e. $nb > na$. This is true because $n > 0$ and $b > a$. □

Common mistake. This is exactly why in Problem 18 we may *not* "add" to the numerator and denominator: adding changes the fraction (a fraction less than 1 gets bigger). Only *multiplying* (or dividing) both by the same non-zero number keeps the fraction the same.

Part c).

Step 6. General form and condition

Fractions equal to $\frac{3}{7}$ have the form $\frac{3k}{7k}$ (k a positive integer). The denominator $7k$ has two digits: $10 \le 7k \le 99$. We have $7\cdot 1 = 7 < 10$, $7\cdot 2 = 14 \ge 10$; $7\cdot 14 = 98 \le 99$, $7\cdot 15 = 105 > 99$. So $k = 2, 3, \ldots, 14$: that is $14 - 2 + 1 = 13$ values. The fractions are $\frac{6}{14}, \frac{9}{21}, \ldots, \frac{42}{98}$.

Answer: a) $n = 7$. b) See the proof. c) 13 fractions.

Problem 44*

Poles, a bicycle and a train (extends Problem 20).

a) An cycles steadily at 18 km/h along a road whose power poles are equally spaced 50 m apart. How many minutes pass from the moment An passes the first pole to the moment he passes the 25th pole?

b) A train travelling at constant speed takes 9 seconds to pass a power pole and 30 seconds to cross a bridge 350 m long (from the moment the front of the train reaches the bridge until the end of the train has just left it). Find the length of the train and its speed in km/h.

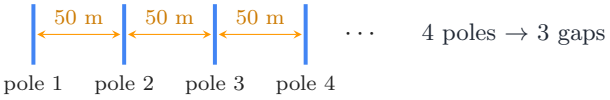
Detailed solution

Understanding the problem. Part a) hides the "fence-post trap": 25 poles make how many gaps? Part b) requires understanding how far a train (an object with length) must travel to "pass" something.

Recall. Distance = speed × time. Converting 1 km/h: 1000 m in 3600 seconds, so 1 km/h $= \frac{1000}{3600} = \frac{5}{18}$ m/s; conversely 1 m/s = 3.6 km/h.

Part a).

Step 1. Count the gaps between poles



Between the first and the 25th pole there are $25 - 1 = 24$ gaps, with total length $24\cdot 50 = 1200$ m.

Step 2. Convert the speed and find the time

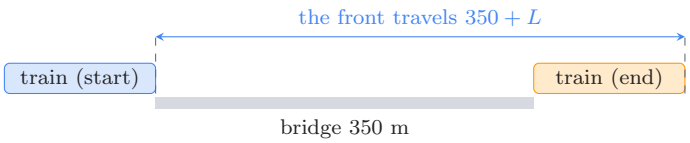
$18 \text{ km/h} = 18 \cdot \frac{5}{18} = 5 \text{ m/s}$. Time: $1200 \div 5 = 240$ seconds $= 240 \div 60 = 4$ minutes.

Common mistake. Counting 25 gaps ($25 \cdot 50 = 1250$ m) gives 250 seconds – wrong. The number of gaps is always one less than the number of poles (counting from the first pole to the last).

Part b).

Step 3. What "passing a pole" and "crossing a bridge" mean

- Passing a pole: from when the front of the train is level with the pole until the end of the train is level with it, the train travels exactly *one train length* L.
- Crossing the bridge: from when the front is at the start of the bridge until the end leaves the far side, the train travels *the bridge length plus the train length*: $350 + L$.



Step 4. Compare the two situations

Crossing the bridge takes 30 seconds, passing the pole takes 9 seconds. The difference $30 - 9 = 21$ seconds is exactly the time to travel the extra 350 m (the bridge length). The speed of the train is

$$v = 350 \div 21 = \frac{350}{21} = \frac{50}{3} \text{ m/s}.$$

Step 5. Find the train length and convert the speed

Train length $L = v \cdot 9 = \frac{50}{3} \cdot 9 = 150$ m. Speed: $\frac{50}{3} \cdot 3.6 = \frac{50 \cdot 3.6}{3} = 60$ km/h.

Common mistake. A common mistake: thinking the train travels only 350 m to cross the bridge (forgetting to add the train length); then the speed would be $350 \div 30 \approx 11.7$ m/s, which is wrong. Follow *one fixed point* on the train (e.g. its front) to know how far the train has travelled.

Check: With $v = \frac{50}{3}$ m/s and $L = 150$ m: passing the pole takes $150 \div \frac{50}{3} = 9$ seconds; crossing the bridge takes $(350 + 150) \div \frac{50}{3} = 500 \cdot \frac{3}{50} = 30$ seconds. Both facts check out.

Answer: a) 4 minutes. b) The train is 150 m long; its speed is 60 km/h.

Problem 45 ★★

Products of digits (extends Problem 21).

a) How many four-digit numbers have digits whose product is 36?

b) Find the smallest positive integer whose digits have product 360.

c) Find the largest positive integer with *all different* digits whose product is 360.

Detailed solution

Understanding the problem. All three parts start by *factorising into primes*, then "sharing" the prime factors among the digits (each digit from 1 to 9). Note: no digit can be 0 (otherwise the product is 0).

Recall. $36 = 2^2 \cdot 3^2$; $360 = 2^3 \cdot 3^2 \cdot 5$. The digits and their factorisations: 1; 2; 3; $4 = 2^2$; 5; $6 = 2 \cdot 3$; 7; $8 = 2^3$; $9 = 3^2$.

Part a).

Step 1. Find the sets of four digits (ignoring order) with product 36

As in Problem 21, list $a \le b \le c \le d$:

- Two digits equal to 1: the other two digits have product 36: $\{1, 1, 4, 9\}$, $\{1, 1, 6, 6\}$.
- Exactly one digit 1: the other three digits (≥ 2) have product 36: $\{1, 2, 2, 9\}$, $\{1, 2, 3, 6\}$, $\{1, 3, 3, 4\}$.
- No digit 1: four digits ≥ 2 with product 36: only $\{2, 2, 3, 3\}$ (since $36 = 2 \cdot 2 \cdot 3 \cdot 3$ cannot be split further).
- Three digits equal to 1: the remaining digit would be 36 – impossible.

Recall. Arranging 4 **digits:** all four different: $4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways. Exactly two equal (the other two different from each other and from them): choose 2 of the 4 positions for the equal pair (6 ways), then place the other two digits in the 2 remaining spots (2 ways): 12 ways. Two equal pairs (like $1, 1, 6, 6$): just choose 2 positions for one pair: 6 ways.

Step 2. Count

Digit set	Type	Arrangements
$\{1, 1, 4, 9\}$	one equal pair	12
$\{1, 1, 6, 6\}$	two equal pairs	6
$\{1, 2, 2, 9\}$	one equal pair	12
$\{1, 2, 3, 6\}$	all different	24
$\{1, 3, 3, 4\}$	one equal pair	12
$\{2, 2, 3, 3\}$	two equal pairs	6
	Total	**72**

Part b).

Step 3. Use as few digits as possible

A number with fewer digits is smaller. One or two digits give a product of at most $9 \cdot 9 = 81 < 360$, so we need at least 3 digits.

Step 4. Find three digits with product 360
360 has the prime factor 5, and the only digit containing the factor 5 is 5 itself. So one digit is 5, and the other two have product $360 \div 5 = 72 = 8 \cdot 9$ (the only way with digits ≤ 9). Digit set: $\{5, 8, 9\}$.

Step 5. Arrange to get the smallest number
Arrange the digits from smallest to largest: 589.

Part c).

Step 6. For the largest number: as many digits as possible
The digits are different, non-zero, and one of them must be 5. Six different digits have product at least $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720 > 360$, so there are at most 5 digits.

Step 7. Find the sets of 5 different digits with product 360
One digit is 5; the other four are different, not 5, with product $72 = 2^3 \cdot 3^2$. Four different digits with a product as small as 72 must include 1 (since $2 \cdot 3 \cdot 4 \cdot 6 = 144 > 72$); the remaining three different digits have product 72: $72 = 2 \cdot 4 \cdot 9 = 3 \cdot 4 \cdot 6$ (while $2 \cdot 6 \cdot 6$ and $3 \cdot 3 \cdot 8$ repeat a digit, and $1 \cdot 8 \cdot 9$ repeats the digit 1). Two sets: $\{1, 2, 4, 5, 9\}$ and $\{1, 3, 4, 5, 6\}$.

Step 8. Arrange to get the largest number
Arrange from largest to smallest: 95421 and 65431. The larger is 95421.

Common mistake. In part c), do not jump to a number with a big first digit such as 985 (product 360): a 5-digit number is always larger than a 3-digit number. Maximise the *number of digits* first, and only then the leading digit.

Check: $5 \cdot 8 \cdot 9 = 360$; $9 \cdot 5 \cdot 4 \cdot 2 \cdot 1 = 360$. A computer search over all numbers with up to 6 digits confirms: there are 72 four-digit numbers with digit product 36; the smallest number with digit product 360 is 589; the largest with all different digits and digit product 360 is 95 421.

Answer: a) 72 numbers. b) 589. c) 95 421.

Problem 46 *

Arithmetic grids (extends Problem 22). In a square grid, the numbers in every row and in every column form arithmetic sequences.

a) A 4×4 grid has these four corner numbers: top left 1, top right 10, bottom left 19, bottom right 37. Fill in the whole grid.

b) Find the sum of all 16 numbers in the grid of part a). Compare it with the sum of the four corners: what do you notice? Explain.

c) Prove that in a 3×3 grid of this kind, the number in the centre always equals the mean of the four corner numbers.

1			10
19			37

Detailed solution

Understanding the problem. Only the four corners are given. The first and last columns (and the first and last rows) each have a known first and last term, so they can be found immediately. Then we use the middle rows or columns.

> **Recall.** Arithmetic sequence with 4 terms: difference $d = (\text{last} - \text{first}) \div 3$. Sum of the 4 terms $= 2 \times (\text{first} + \text{last})$ (pair term 1 with term 4, term 2 with term 3: both pairs have the same sum). With 3 terms: middle term = mean of the two outer terms.

Part a).

Step 1. First and last rows

Row 1: from 1 to 10, $d = (10 - 1) \div 3 = 3$: $1, 4, 7, 10$.
Row 4: from 19 to 37, $d = (37 - 19) \div 3 = 6$: $19, 25, 31, 37$.

Step 2. The four columns (first and last terms of each are known)

Column 1: $1 \to 19$, $d = 6$: $1, 7, 13, 19$. Column 2: $4 \to 25$, $d = 7$: $4, 11, 18, 25$.
Column 3: $7 \to 31$, $d = 8$: $7, 15, 23, 31$. Column 4: $10 \to 37$, $d = 9$: $10, 19, 28, 37$.

Step 3. Check the middle rows

Row 2: $7, 11, 15, 19$ (difference 4); row 3: $13, 18, 23, 28$ (difference 5). The grid satisfies every condition:

1	4	7	10	row sum: 22
7	11	15	19	row sum: 52
13	18	23	28	row sum: 82
19	25	31	37	row sum: 112

Part b).

Step 4. Compute the sum

Sum of the rows: $22 + 52 + 82 + 112 = 268$. Sum of the corners: $1 + 10 + 19 + 37 = 67$. We see that $268 = 4 \cdot 67$: **the sum of the whole grid is 4 times the sum of the corners.**

Step 5. Explanation (true for every such 4×4 grid)

Each row is an arithmetic sequence of 4 terms, so each row sum $= 2 \times$(first term + last term of the row). Adding the 4 rows:

$$\text{Grid sum} = 2 \times (\text{sum of column 1} + \text{sum of column 4}).$$

Columns 1 and 4 are also arithmetic sequences of 4 terms: sum of column $1 = 2 \times (1 + 19)$, sum of column $4 = 2 \times (10 + 37)$. So

$$\text{Grid sum} = 2 \times \big(2 \cdot (1 + 19) + 2 \cdot (10 + 37)\big) = 4 \times (1 + 10 + 19 + 37) = 4 \times 67 = 268.$$

Part c).

Step 6. Name the cells of the 3×3 grid

a	$\cdot$	b
u	x	v
c	$\cdot$	d

The corners are a, b, c, d; the centre is x; u and v are the middle cells of the left and right columns.

Step 7. Use "middle term = mean of the outer terms"

The left column a, u, c is arithmetic, so $u = \dfrac{a+c}{2}$. Right column: $v = \dfrac{b+d}{2}$. The middle row u, x, v is arithmetic, so

$$x = \frac{u+v}{2} = \frac{\frac{a+c}{2} + \frac{b+d}{2}}{2} = \frac{a+b+c+d}{4}.$$

This is exactly the mean of the four corner numbers.

Why is it true? Why does an arithmetic sequence p, q, r satisfy $q = \dfrac{p+r}{2}$? Because $q - p = r - q$ (both equal the difference d), so $2q = p + r$. □

Check: Apply part c) to the top-left 3×3 block of the grid in part a): corners $1, 7, 13, 23$, mean $\dfrac{1+7+13+23}{4} = 11$, exactly the centre 11.

Answer: a) See the grid. b) Sum $268 = 4 \times 67$ (4 times the sum of the corners). c) See the proof.

Problem 47 **

"Rotating" numbers (extends Problem 23). In each part, every letter represents a digit, and $A \neq 0$.

a) Find the six-digit number $\overline{2ABCDE}$ such that $\overline{2ABCDE} \times 3 = \overline{ABCDE2}$.

b) Find all digits d (from 1 to 9) for which there is a six-digit number $\overline{dABCDE}$ with $\overline{dABCDE} \times 3 = \overline{ABCDEd}$.

c) Prove that there is no six-digit number $\overline{1ABCDE}$ with $\overline{1ABCDE} \times 2 = \overline{ABCDE1}$.

Detailed solution

Understanding the problem. All three parts are about "moving the first digit to the end". We reuse Method 1 of Problem 23: treat the block $\overline{ABCDE}$ as one number N and set up an equation. This time we do it once for every leading digit d.

Recall. With $N = \overline{ABCDE}$: $\overline{dABCDE} = d \cdot 100\,000 + N$ and $\overline{ABCDEd} = 10 \cdot N + d$. Since $A \neq 0$, N has exactly five digits: $10\,000 \leq N \leq 99\,999$.

Step 1. The general equation for parts a) and b)

$$3 \cdot (d \cdot 100\,000 + N) = 10N + d \implies 300\,000\,d + 3N = 10N + d \implies 299\,999\,d = 7N.$$

Since $299\,999 = 7 \cdot 42\,857$, dividing both sides by 7 gives

$$N = 42\,857 \cdot d.$$

Step 2. Part a): $d = 2$
$N = 42\,857 \cdot 2 = 85\,714$. The number is 285 714.

Step 3. Part b): when does N have five digits?
$N = 42\,857\,d \le 99\,999$. For $d = 1$: $N = 42\,857$ (five digits, fine). For $d = 2$: $N = 85\,714$ (fine). For $d = 3$: $N = 128\,571$ has six digits – too big to be the block $\overline{ABCDE}$. For $d \ge 3$, N is even bigger. So only $d = 1$ and $d = 2$ work, giving the two numbers 142 857 and 285 714.

ThinkAlgo Tip. These two numbers use the same digits $1, 4, 2, 8, 5, 7$ arranged in a circle. That is no accident: $\frac{1}{7} = 0.\overline{142857}$ and $\frac{2}{7} = 0.\overline{285714}$. Multiplying by 3 corresponds to "rotating" the circle of digits.

Step 4. Part c): set up the equation
With $N = \overline{ABCDE}$:

$$2 \cdot (100\,000 + N) = 10N + 1 \implies 200\,000 + 2N = 10N + 1 \implies 199\,999 = 8N.$$

Step 5. Part c): an odd/even contradiction
The right-hand side $8N$ is even (since 8 is even). The left-hand side 199 999 is odd. An odd number cannot equal an even number. So no such number exists.

Why is it true? You can see it immediately without an equation: $\overline{1ABCDE} \times 2$ is even, so its last digit must be even; but $\overline{ABCDE1}$ ends in 1 (odd). Contradiction. □

Common mistake. In part b), many students stop after finding $d = 1$ (the answer of Problem 23) and forget $d = 2$; or they accept $d = 3, 4, \ldots$ without checking whether N has more than five digits. Always check the condition "N has exactly five digits".

Check: $142\,857 \times 3 = 428\,571$; $285\,714 \times 3 = 857\,142$. In both cases the first digit has indeed moved to the end.

Answer: a) 285 714. b) $d = 1$ or $d = 2$ (the numbers 142 857 and 285 714). c) See the proof.

Problem 48 ★★

A frog on the number line (extends Problem 24). A frog sits at the point 0 on the number line. On its k-th jump ($k = 1, 2, 3, \ldots$), the frog jumps exactly k units, to the left or to the right as it likes.

a) After exactly 10 jumps, can the frog be back at 0? Why?

b) After exactly 10 jumps, which points can the frog be at? How many such points are there?

c) What is the smallest number of jumps the frog needs to reach the point 20?

Detailed solution

Understanding the problem. The frog's position after n jumps is $\pm 1 \pm 2 \pm 3 \cdots \pm n$ (a $+$ sign for a jump to the right, a $-$ sign for a jump to the left). The idea is the same as in Problem 24: start from "all plus" and see how switching signs changes the result.

Recall. $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$ (Gauss's formula). For example $1 + 2 + \cdots + 10 = \dfrac{10 \cdot 11}{2} = 55$.

Step 1. The "switching" rule

If the frog always jumps right, after 10 jumps it is at 55. Switching one jump k from right to left lowers the position by $2k$ (it loses k and goes back another k). So if the jumps to the left add up to s, the final position is

$$55 - 2s.$$

Part a).

Step 2. Odd or even

$55 - 2s$ is always *odd* (odd minus even). The point 0 is even, so the frog **cannot** be back at 0 after 10 jumps.

Part b).

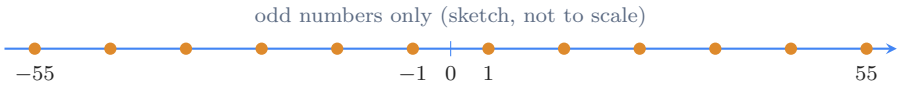
Step 3. Which values can s take?

s is the sum of a group (possibly empty) of different numbers from $\{1, 2, \ldots, 10\}$, so $0 \le s \le 55$. We show that *every* integer s from 0 to 55 can be made.

Why is it true? Take the largest numbers one by one: $10, 9, 8, \ldots$ The totals 0, 10, $10 + 9 = 19$, $19 + 8 = 27$, $27 + 7 = 34$, $34 + 6 = 40$, $40 + 5 = 45$, $45 + 4 = 49$, $49 + 3 = 52$, $52 + 2 = 54$, $54 + 1 = 55$ cut the interval $[0, 55]$ into pieces. A number s between two consecutive marks, e.g. $27 < s \le 34$, is written as $s = 10 + 9 + 8 + (s - 27)$ with $s - 27$ from 1 to 7 – a number not yet used. The same works for every other piece (for $0 < s \le 10$ just take the number s). So every s from 0 to 55 can be made. □

Step 4. The final positions

For $s = 0, 1, 2, \ldots, 55$, the position $55 - 2s$ takes the values $55, 53, 51, \ldots, -53, -55$: that is, **every odd number from -55 to 55**. Counting: from -55 to 55 there are 28 positive odd numbers $(1, 3, \ldots, 55)$ and 28 negative ones, 56 points altogether (the same as the number of values of s: 56).



Part c).

Step 5. Two necessary conditions

After n jumps, let $T = 1 + 2 + \cdots + n$. The position is $T - 2s$, so: (1) $T \ge 20$ (the frog cannot get farther than T); (2) $T - 20$ must be even, i.e. T must be even like 20.

Step 6. Try n one at a time

n	1	2	3	4	5	6	7
T	1	3	6	10	15	21	28
Reach 20?	no ($T < 20$)					no (T odd)	**possible**

Step 7. Show a way with $n = 7$
We need $28 - 2s = 20$, i.e. $s = 4$: just jump left on the 4th jump:

$$1 + 2 + 3 - 4 + 5 + 6 + 7 = 20.$$

So the smallest number of jumps is 7.

Common mistake. Checking only "$T \geq 20$" is not enough: for $n = 6$, $T = 21 \geq 20$, but the position is always odd and can never be 20. This is exactly the odd/even idea of Problem 24.

Check: $1 + 2 + 3 - 4 + 5 + 6 + 7 = 20$. For $n = 6$ the position is always $21 - 2s$, an odd number, never 20. A computer search of all jump patterns also confirms: after 10 jumps there are exactly 56 positions, not including 0; the smallest number of jumps to reach 20 is 7.

Answer: a) No (the position is always odd). b) Every odd number from -55 to 55: 56 points. c) 7 jumps.

Problem 49 *

Numbers containing the block 2025 (extends Problem 28). Consider the six-digit numbers that contain the four digits $2, 0, 2, 5$ *together and in this order* (for example: 720 251, 202 513, 132 025). How many of these numbers are divisible by 45?

Detailed solution

Understanding the problem. The number consists of the block 2025 plus two more digits a, b in different positions. "Divisible by 45" should be split into two conditions that are easier to check.

Recall. $45 = 5 \cdot 9$, and 5 and 9 have no common divisor other than 1. So a number is divisible by 45 exactly when it is divisible by both 5 and 9. **Divisible by 5:** the last digit is 0 or 5. **Divisible by 9:** the digit sum is divisible by 9.

Step 1. The condition for divisibility by 9
The digits of the block add up to $2 + 0 + 2 + 5 = 9$. The whole number has digit sum $9 + a + b$, which is divisible by 9 when $a + b$ is divisible by 9, i.e. $a + b \in \{0,\ 9,\ 18\}$.

Step 2. The three positions of the block, case by case

- **Form** $\overline{2025ab}$ (the last digit is b, so $b \in \{0, 5\}$; a is any digit): $b = 0$: $a \in \{0, 9\} \to 202\,500$, $202\,590$. $b = 5$: $a + 5 \in \{9, 18\} \to a = 4 \to 202\,545$. **3 numbers.**
- **Form** $\overline{a2025b}$ ($a \neq 0$, $b \in \{0, 5\}$): $b = 0$: $a = 9 \to 920\,250$. $b = 5$: $a = 4 \to 420\,255$. **2 numbers.**

- **Form** $\overline{ab2025}$ (ends in 5 – always divisible by 5; $a \neq 0$): $a+b=9$: $a=1,2,\ldots,9$ (with $b=9-a$) $\to$ 9 numbers: 182 025, 272 025, …, 902 025. $a+b=18$: $a=b=9 \to 992\,025$. ($a+b=0$ is impossible because $a \neq 0$.) 10 **numbers.**

Step 3. Check for double counting

A number would be counted twice if the block 2025 appeared twice among its six digits, i.e. two overlapping blocks. Shifting the block by 1 place: "025" would have to match "202" – false; shifting by 2 places: "25" would have to match "20" – false. So no number is counted twice.

Step 4. Add up

$3+2+10=15$ numbers.

Common mistake. (1) Do not forget the case $a+b=0$ in the form $\overline{2025ab}$ (the number 202 500). (2) In the forms $\overline{a2025b}$ and $\overline{ab2025}$, the first digit a cannot be 0. (3) Check *both* conditions, divisibility by 5 and by 9, not just one.

Check: Example 992 025: it ends in 5, and its digit sum $9+9+2+0+2+5=27$ is divisible by 9, so it is divisible by 45 ($992\,025 = 45 \cdot 22\,045$). A computer search over all six-digit numbers also gives exactly 15.

Answer: 15 **numbers.**

Problem 50 ★★

Mean and median (extends Problem 30). A list of five positive integers (repeats allowed) has mean 12 and median 10.

a) What is the largest possible value of the largest number in the list?

b) What is the smallest possible value of the largest number in the list?

c) Answer both questions again if the five numbers must all be different.

Detailed solution

Understanding the problem. Arrange the five numbers in increasing order: $a \le b \le c \le d \le e$. The median is the middle number, so $c=10$. The mean is 12, so the sum is $5 \cdot 12 = 60$. We want to make e as large or as small as possible. This is an "optimisation" problem: we must both *prove* that a limit cannot be passed and *give an example* that reaches exactly that limit.

Recall. Mean = sum ÷ how many numbers, so sum $= 12 \cdot 5 = 60$. The median of five numbers (in increasing order) is the third one.

Step 1. The key equation

$$a+b+10+d+e=60 \quad \Longrightarrow \quad a+b+d+e=50, \qquad a \le b \le 10 \le d \le e.$$

a	b	10	d	e	sum = 60, median = 10

Part a): make e as large as possible.

Step 2. Upper limit
For e to be large, the other numbers must be as small as possible: $a \geq 1$, $b \geq 1$, $d \geq 10$. Hence $e = 50 - a - b - d \leq 50 - 1 - 1 - 10 = 38$.

Step 3. An example that reaches it
The list $1, 1, 10, 10, 38$: sum 60, median 10. So the largest possible e is 38.

Part b): make e as small as possible.

Step 4. Lower limit
For e to be small, a and b must be as large as possible, but $a \leq b \leq 10$, so $a + b \leq 20$. Hence $d + e = 50 - (a + b) \geq 30$. Since $d \leq e$, $2e \geq d + e \geq 30$, so $e \geq 15$.

Step 5. An example that reaches it
The list $10, 10, 10, 15, 15$: sum 60, median 10. So the smallest possible e is 15.

Part c): five different numbers ($a < b < 10 < d < e$).

Step 6. Largest e
As small as possible: $a \geq 1$, $b \geq 2$, $d \geq 11$. So $e \leq 50 - 1 - 2 - 11 = 36$. Example: $1, 2, 10, 11, 36$ (sum 60). So the largest e is 36.

Step 7. Smallest e
As large as possible: $b \leq 9$, $a \leq 8$, so $a + b \leq 17$ and $d + e \geq 50 - 17 = 33$. Since $d < e$, $d \leq e - 1$, so $2e - 1 \geq d + e \geq 33$, i.e. $e \geq 17$. Example: $8, 9, 10, 16, 17$ (sum 60). So the smallest e is 17.

Common mistake. A "largest/smallest" problem needs *two* parts: (1) a proof that the limit cannot be passed, and (2) a concrete example reaching exactly that limit. An example alone may not be the best; a limit alone may not be reachable.

Note for parents. Let your child share 60 sweets among 5 "boxes" kept in order, with the middle box always holding 10. Ask: to make the last box as full as possible, should the other boxes hold few or many? And to make the last box as small as possible? This game helps children "feel" optimisation reasoning.

Check: $1 + 1 + 10 + 10 + 38 = 60$; $10 + 10 + 10 + 15 + 15 = 60$; $1 + 2 + 10 + 11 + 36 = 60$; $8 + 9 + 10 + 16 + 17 = 60$; all have median 10. A computer search over all lists also confirms the values 38, 15, 36, 17.

Answer: a) 38. b) 15. c) Largest 36, smallest 17.